

Exercises**Level - 1****(Practice Questions Based on Fundamentals)****FACTORIAL NOTATION**

1. Find x , if $\frac{x}{5!} + \frac{x}{6!} = \frac{1}{7!}$.
2. Simplify: $\frac{(2n)!}{n!}$.
3. Simplify: $\frac{3 \cdot 6 \cdot 9 \cdot 12 \dots (3n-3)3n}{3^n}$
4. Find the unit digit of $1! + 2! + 3! + 4! + \dots + (10)!$.
5. If $N = a! + b! + c! + d! + e!$ be a two-digit number, find the value of N , where $a, b, c, d, e \in \mathbb{I}$.
6. If the product of factorials of n consecutive positive integers be a single-digit number, find the maximum value of n .

7. If the sum of the factorials of N consecutive natural numbers be a three-digit number, find the maximum value of N .
8. If the unit digit of $N = a! + b! + c!$ is 9, where a, b and c are positive integers, find the product of $\{a! \times b! \times c!\}$.
9. If N be the sum of factorials of all the prime numbers less than 100, find the last two digits of N .
10. A three-digit number xyz such that $x + y + z = 25$. Find the sum of all possible values of $x!y!z!$.
11. Find the sum of $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + \dots + n \cdot n!$.
12. Find the sum of
$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \dots + \frac{99}{100!}.$$

EXPONENT OF A PRIME P IN $(N)!$

13. Find the exponent of 3 in $(10)!$
14. Find the exponent of 5 in $(50)!$
15. Find the highest power of 10 in $50! + 60! + 70!$
16. Find the highest power of 10 in $50! \times 60! \times 70!$
17. Find the highest power of 10 in $10! + 20! + 30! + 40! + \dots + (2020)!$
18. Prove that $33!$ is divisible by 2^{19} .
19. Find the number of zeroes at the end of $(100)!$.
20. If the highest power of 10 in $N!$ be 16, find the highest power of 10 in $(N + 1)!$
21. If $m(n)!$ be completely divisible by 5^{11} , where m is a single-digit number, find the minimum value of n .
22. If N be the product of the first 100 multiples of 5, find the highest power of 10 in N .

FUNDAMENTAL PRINCIPLE OF COUNTING

23. Suppose, there are 3 different ways you can come from your home to Sakchi and 4 different ways you can come from Sakchi to MIT-JEE Institute. In how many ways you can come from your home to MIT-JEE Institute.
24. There are 5 doors in your classroom. In how many ways a student can enter the classroom and leave by different door?
25. In how many different ways can three students queue up at a bus stop?
26. 5 students compete in a race. In how many ways first three prizes be given?
27. A question paper consists of two sections A and B respectively. There are 5 and 7 questions in section A and B , respectively. In how many ways a student can attempt the questions?
28. A student has 5 different books. In how many ways he can arrange the books in a shelf of his almirah?
29. How many distinct integers are there in between 100 and 1000?
30. Find the number of terms in the product $(a + b)(c + d + e)(f + g + h + i)(j + k + l + m + n)$.

31. Find the total number of 3-digit numbers which can be made from the digits 1, 2, 3, 4 and 5.
32. Find the number of positive integral solutions of $x + y = 10$.
33. A question paper has two sections A and B respectively. Section A has 7 questions whereas section B has 6 questions, respectively. In how many ways a student can attempt for a single question either from section A or section B ?
34. In a class, there are 15 boys and 10 girls. In how many ways your class teacher can select a class monitor?
35. How many 3-digit numbers can be made by the digits either 3 or 4 or 5?
36. How many distinct positive numbers can be made by the digits 1, 2, 3, 4, and 5?
37. How many three-digit distinct numbers can be made by the digits 1, 2, 3, 4 and 5 in which all are either even or odd numbers?
38. Find the number of positive integral solutions of $a + b + c = 6$.

PERMUTATIONS

39. Suppose 8 people enter an event in a swim meet. In how many ways could the gold, silver and bronze medals be awarded?
40. How many integers between 100 and 999 (inclusive) consist of odd digits?
41. How many 3-digits even numbers can be formed by using the digits 1, 2, 3, 4 and 5?
42. Find the number of ways 6 flags of different colors are to be used to form a signal of 4 flags.
43. A flag containing 4 strips of different colours to be designed. If 6 colours can be used, find the number of possible flags.
44. In an experiment on social interaction, 6 people can sit in 6 seats in a row. In how many ways this can be done?
45. In how many ways 3 men stay in a hotel room, if 8 rooms are available?
46. In a club with 15 members, how many ways 3 officers—president, vice-president and treasurer can be chosen?
47. Find the number of permutations of the letters of the word TABLE.
48. If a be the permutations of the letters of the word MAIN, b be the permutations of the letters of the word EXAM, c be the permutations of the letters of the word LOVE, find the value of $a + b + c + 10$.
49. How many 4-digit numbers can be made by the digits 1, 2, 3, 4 and 5, where repetition is not allowed?
50. How many 4-digit distinct odd numbers can be made from the digits 3, 4, 5, 6 and 7?
51. How many numbers lying in between 100 and 1000 can be made from the digits 0, 1, 2, 3, 4 and 5, where repetitions of the digits is not allowed?

52. How many numbers in between 300 and 1000 can be made from the digits 0, 2, 3, 4, 5 and 6, where repetition is not allowed?
53. How many 4-digit even numbers can be made by the digits 0, 2, 3, 4, 5, 6 and 7?
54. How many 5-digit even numbers can be formed from the digits 1, 2, 3, 4 and 5?
55. How many 5-digit odd numbers divisible by 5 can be made by the odd digits?
56. How many 5-digit even numbers divisible by 5 can be made by the digits 0, 1, 2, 3, 4, 5, 6, 7 and 8?
57. How many distinct positive numbers can be made from the digits 0, 1, 2, 3 and 4, where repetition is not allowed?
58. How many numbers of 4-digits greater than 2000 can be made from the digits 0, 1, 2, 3, 4 and 5, where the repetition is not allowed?
59. Suppose two sets A and B consist of 4 and 7 elements, respectively. How many one one functions you can make from A to B ?
60. Consider two sets A and B have 5 elements respectively. How many one one onto functions you can make from A to B ?
61. Suppose two sets A and B have 4 elements, respectively. How many inverse functions are exist from A to B ?
62. How many onto functions can be defined from a set A to a set B , if the sets A and B are consisting of 5 and 3 elements, respectively.
63. Find the number of onto functions between two sets A and B having 5 and 4 elements respectively.

SUM OF NUMBERS

64. Find the sum of the digits in the unit place of all numbers made by the digits 2, 3, 4 and 5.
65. Find the sum of all four-digit numbers made by the digits 1, 2, 3 and 4.
66. Find the sum of all four-digit numbers that can be made from the digits 0, 1, 2 and 3.
67. Find the sum of all 4-digit numbers that can be made by the digits 2, 3, 3 and 4.

PERMUTATION WITH REPETITIONS

68. How many different outcomes are there in an experiment consisting of n tosses of a coin?
69. In how many ways 4 delegates can be put in 5 hotels, if there is no restriction?
70. In how many ways 3 friends can be put up in a 5 hotel of a town, if
 - (i) there is no restriction?
 - (ii) no two friends can stay together?
71. In how many ways 5 rings can be worn on the four fingers of one hand?
72. If m is the number of ways, a child put 3 marbles in his 4 pockets and n is the number of ways 3-digit distinct numbers can be made from the digits 1, 2, 3, 4 and 5, find the value of $m + n + 10$.

73. In how many ways 5 prizes can be given to 10 students, if each student can get any number of prizes?
74. How many ATM pin codes can be made by the digits of a bank?
75. How many 3 letters code words can be made from the word I N T E X (where repetition is allowed)?
76. In how many ways can a ten questions multiple choice answered in JEE Main examination, if there are four choices a , b , c and d to each question?
77. A student of MIIT-JEE Institute wants to appear in JEE advanced mock examination, which contains 10 multiple choice questions. Each question has four choices a , b , c and d , respectively and more than one correct answers are given. In how many ways the student can give the answer?

PERMUTATIONS OF ALIKE OBJECTS

78. Find the number of permutations of the letters of the word IIT.
79. In how many ways can you permute the letters of the word AIEEE?
80. In how many ways can you permute the letters of the word MATHEMATICS?
81. In how many ways can you permute the letters of the word CONSTITUTION?
82. How many seven-digit numbers can be formed with the digits 1, 2, 2, 2, 3, 3 and 5?
83. Find the sum of all the numbers that can be made by the digits 2, 3, 3, 4, 4 and 4?
84. If m be permutation of the letters of the word JEE-MAIN and n be the permutations of the letters of the word IIT, find the value of $m + n + 10$.
85. Find the number of ways in which 10 students can be arranged in a row such that A is always ahead of B .
86. Find the number of ways in which four particular persons A , B , C , D and six more persons can stand in a queue so that A always stands before B , B before C and C before D .
87. Find the number of permutations of the letters a , b , c and d such that b does not follow a , c does not follow b and d does not follow c .

RESTRICTED PERMUTATIONS

88. In how many ways, 10 boys can be seated in a row so that 2 boys always sit together?
89. In how many ways, 6 boys and 5 girls can be seated in a row so that all boys and girls are sit together?
90. Find the number of ways in which the letters of the word FRACTION be arranged so that all vowels are together.
91. Find the number of ways in which the letters of the word EQUATION be arranged so that all vowels are together.

92. Find the number of ways the letters of the word LAUGH can be arranged so that all vowels and consonants are together.
93. How many different words can be made by the letters of the word UNIVERSITY, so that all vowels are together?
94. In how many ways, 10 students can be seated in a row so that three students A , B and C are always together and C is always ahead of A and B ?
95. Find the number of words can be made by the letters of the word ALGEBRA so that all vowels do not come together.
96. Find the number of words that can be made out of the letters of the word INTEGER so that I and N are never come together.
97. Find the number of arrangements that can be made out of the letters of the word SUCCESS so that all S s do not come together.
98. Find the number of arrangements of the letters of the word BANANA in which two N s do not appear adjacently.
99. Find the number of arrangements of the letters of the word PARALLEL so that all L s do not come together but all A s come together.
100. How many words, can be made by the letters of the word INTERMEDIATE so that no vowels in between two consonants?
101. In how many ways, can the letters of the word NINETEEN be arranged so that neither all the vowels nor all the consonants are together?

RANK OF A WORD IN DICTIONARY

102. Find the rank of the word TOUGH in the English dictionary.
103. Find the rank of the word IIT in the English dictionary.
104. Find the rank of the word AIEEE in the English dictionary.
105. Find the rank of the word ANNA in the English dictionary.
106. Find the rank of the word PATNA in the English dictionary.
107. Find the rank of the word SURITI according to the English dictionary.
108. Find the rank of the word SANIA in the English dictionary.

GAP METHOD

109. Find the number of ways in which the letters of the word FRACTION be arranged so that no two vowels are together.
110. Find the number of ways in which the letters of the word TRIANGLE be arranged so that no two vowels are together.

111. All the letters of the word EAMCET are arranged in all possible ways. Find the number of such arrangements in which no two vowels are adjacent to each other.
112. In how many ways, letters of the word NINETEEN be arranged so that no two N s are together?
113. In how many ways, letters of the word NINETEEN be arranged so that no two vowels as well as no two consonants are together?
114. In how many ways, 5 girls and 6 boys in a class can be seated in a row so that the boys and girls are alternate?
115. In how many ways, 5 boys and 5 girls are seated in a row so that they are alternate?
116. How many different nine-digit numbers can be formed from the number 223355888 by re-arranging its digit so that odd digits occupy even positions?
117. How many words can be made from the letters of the word MATHEMATICS without changing the relative order of vowels and consonants?
118. How many words can be made from the letters of the word ACCIDENT without changing the relative order of vowels?
119. How many words can be made from the letters of the word INEFFECTIVE keeping the position of each vowel is fixed?

CIRCULAR PERMUTATIONS

120. In how many ways, 6 Indians and 5 Englishmen can be seated in a round table if
 - (i) there is no restriction?
 - (ii) all the 5 Englishmen sit together?
 - (iii) all the 5 Englishmen do not sit together?
 - (iv) no two Englishmen sit together?
121. In how many ways, can we arrange 6 different flowers in a circle?
122. In how many ways, can 50 different pearls be arranged to form a necklace?
123. In how many ways, 3 ladies and 5 gentlemen arrange themselves around a round table so that every gentleman may have at least one lady?
124. In how many ways, 20 persons can sit around a round table in such a way that there is exactly one person between A and B ?
125. In how many ways, 7 boys and 8 girls can be seated around a circular table?
126. In how many ways, 7 boys and 8 girls can be seated around a circular table having 15 chairs numbered from 1 to 15?
127. In how many ways, 10 students can be seated around a circular table in which 3 students A , B and C are always together and B is always between A and C ?
128. In how many ways, 20 persons can be seated around two circular tables having 10 seats each?
129. In how many ways, 20 persons be seated around a round table if there are 10 seats are available there?

130. In how many ways, 20 different pearls of two colors can be set alternately on a necklace, there being 10 pearls of each color?
131. How many necklace of 20 beads each can be made from 10 beads of various colors?

COMBINATIONS

132. In how many ways, 7 different books be given to 2 students—one gets 4 books and other gets 3 books?
133. In how many ways, Master Rohan can invites 4 friends out of his 10 friends in a dinner?
134. In how many ways, I can select the best 5 mathematics students out of 12 students for an Olympiad examination?
135. In how many ways, a team of 3 boys and 4 girls can be selected out of 7 boys and 9 girls?
136. In how many ways, 5 vowels and 5 consonants can be selected from 26 letters in the English alphabet?
137. If a bus conductor has 190 different tickets, find the number of stoppage that the bus has consider only one way journey?
138. On a new year day, every student of a class sends a card to every other student. The postman delivers 600 cards. How many students are there in the class?
139. A man has 10 children to take them in a circus. He takes 4 of them at a time to the circus as often as he can without taking the same 4 children together more than once. How many times will he has to go to circus?
140. In an election, three wards of a town are canvassed by 4, 5 and 8 men, respectively. If there are 20 volunteers, in how many ways can they have allotted to different wards?
141. In how many ways, 4 cards are choosing from a pack of 52 cards in which all are of the same suit?
142. In how many ways, 4 cards are choosing from a pack of 52 cards in which all are of different suits?
143. Out of 6 gentlemen and 4 ladies, a committee of 5 is to be formed. In how many ways can this be done so as to include at least one lady in each committee?
144. Find the domain of the function $f(x) = {}^{2x-5}C_{x-2}$.
145. Find the range of the function $f(x) = {}^{3x-7}C_{x-1}$.
146. If ${}^{10}C_r = {}^{10}C_{r+2}$, find r .
147. If ${}^nC_{n-4} = 15$, find the value of n .
148. If ${}^{15}C_r : {}^{15}C_{r-1} = 11 : 5$, find r .
149. Find the value of ${}^{20}C_{13} + {}^{20}C_{15} + {}^{20}C_9 - {}^{20}C_7 - {}^{20}C_5 - {}^{20}C_{11}$.
150. If ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^nC_3$, prove that $n > 7$.
151. Find the value of ${}^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3$.
152. If ${}^nC_{r-1} = 36$, ${}^nC_r = 84$ and ${}^nC_{r+1} = 126$, find n and r .
153. Solve for x : ${}^{x+3}P_3 = {}^{x+2}C_3 + 20$.
154. If $m = {}^nC_2$, find the value of mC_2 .
155. If $\frac{1}{4C_n} = \frac{1}{5C_n} + \frac{1}{6C_n}$, find n .

RESTRICTED COMBINATIONS

156. In how many ways, can we select the best 20 students for JEE-Advanced examination from a batch of 100 students, if
- two particular students are always chosen?
 - five particular students are never chosen?
157. A group of students consists of 4 boys and 5 girls. Find the number of ways of selecting a team of at least 3 boys and 4 girls.
158. How many ways are there choosing a hand of 6 cards containing an ace and a king of the same suit from a pack of 52 cards?
159. 10 cards are taken out from a well-shuffle pack of 52 cards. In how many ways of choosing a hand of 10 cards consists of at least one ace?
160. A bag contains 5 black and 5 red balls, all balls being different. Find the number of ways in which 2 black and 3 red balls can be selected.
161. Out of 6 gentleman and 4 ladies, a committee of 5 is to be formed. In how many ways, can this be done so as to include at least one lady in each committee?
162. From 6 boys and 7 girls, a committee of 5 is to be formed so as to include at least one girl. Find the number of ways this can be done?
163. A candidate is required to answer 7 questions out of 12 questions, which are divided into two groups containing 6 questions each. He is not permitted to attempt more than 5 from either group. In how many different ways, can be chosen the seven questions?
164. In an examination, the question paper contains three different sections A , B and C containing 4, 5 and 6 questions respectively. In how many ways, a candidate can make a selection of 7 questions, selecting at least two questions from each section.
165. In an election for 3 seats, there are 6 candidates. A voter cannot vote for more than 3 candidates. In how many ways can he vote?

COMBINATIONS FROM DISTINCT OBJECTS

166. Master Amit has 8 friends. In how many ways he can invite one or more of them in his birthday?
167. In how many ways, Master Roshan can give answer at least one question out of 10 questions in his class test?
168. In an examination, a student will pass if he/she passes all 5 subjects in his class. In how many ways he/she will fail the examination?
169. A questions paper consists of two sections having 3 and 5 questions respectively. One question from each section is compulsory. In how many ways can a student select the question?
170. A bag contains 2 white and 3 red cubes, all of different sizes. In how many ways can 3 cubes be selected from the bag if

- (i) at least one cube is white?
 (ii) at least one cube is red?
171. There are n students in a class and a group of three is taken by class teacher to zoo. The same set of three children is not taken more than once. It is found that he goes to zoo 84 times more than the visits of a particular child, find the value of n .
172. In a club election, the number of contestants is one more than the maximum number of the candidates which a voter can vote for. If the total number of ways a voter can vote is 126, find the number of contestants.
173. In a chess tournament, the participants were to play one game with one another. Two players fell ill having played 6 games each and without playing with each other. If the total number of games played is 117, find the number of participants at the beginning.
174. A person is permitted to select at least 1 and atmost n coins from a collection of $(2n + 1)$ distinct coins. If the total number of ways in which it can be done is 255, find the value of n .
175. A student is allowed to select atmost n books from a $(2n + 1)$ books. If the total number ways he can select at least one book is 63, find the value of n .

COMBINATIONS FROM IDENTICAL OBJECTS

176. In how many ways, Master Hemant can select 10 red balls out of 20 red balls?
177. In how many ways, Master Kalicharan can select zero or more balls from 12 green balls?
178. In how many ways, Master Bheem can select at least 1 ball out of 10 red balls, 15 green balls, 12 blue balls and 5 black balls?
179. Out of 6 apples, 5 mangoes and 4 bananas, how many selections of fruits can be made?
180. Out of 6 apples, 5 mangoes and 7 bananas, how many selections of fruits can be made such that at least one fruit of each type is always included?
181. Out of 10 apples, 6 mangoes and 4 bananas, how many selections of fruits can be made such that at least one mango is always included?
182. In a fruit basket, 4 mangoes, 5 bananas and 3 different types of fruits are kept. In how many ways one can select fruits from this fruit basket?
183. Find the number of ways in which one or more letters be selected from the letter of the word ABRACADABRA.
184. Find the number of ways in which one or more letters be selected from the letter of the word MATHEMATICS.
185. Find the number of ways in which one or more letters be selected from the letters of the sequence DADDY DID A DEADLY DEED.
186. Find the number of ways in which a selection of 4 letters can be made from the letters of the word PROPORTION.
187. Find the number of ways in which an arrangement of 4 letters can be made from the letters of the word PROPORTION.
188. Find the number of ways in which a selection of 4 letters can be made from the letters of the word MATHEMATICS.
189. Find the number of ways in which an arrangement of 4 letters can be made from the letters of the word MATHEMATICS.
190. Find the number of ways in which a selection of 4 letters can be made from the letters of the word PASSPORT.
191. Find the number of ways in which an arrangement of 4 letters can be made from the letters of the word PASSPORT.
192. Find the number of ways in which we can select 5 letters of the word INTERNATIONAL.
193. Find the number of ways in which an arrangement of 5 letters can be made from the letters of the word INTERNATIONAL.

DIVISOR OF A GIVEN NATURAL NUMBER

194. If $\alpha = 10800$, find the
- total number of divisors of α
 - the number of even divisors of α
 - the number of divisors α of the form $(4m + 2)$
 - the number of divisors α , which are multiple of 15.
195. In how many ways, the number 18900 can be split into two factors which are relatively prime or co-prime?
196. Find the number of ordered pairs (p, q) such that LCM of p and q is 2520.
197. Find the sum of all divisors of 2520.
198. Find the sum of all odd divisors of 360.
199. Find the sum of all even divisors of 38808.
200. Find the sum of all factors of 7200 that ends with 5.
201. How many number less than 10000 has only 3 factors?
202. In how many ways 8100 can be resolved into product of two factors?
203. In how many ways 10800 can be resolved as a product of two factors?

DISTRIBUTION INTO GROUPS

204. In how many ways 4 different things can be divided P, Q, R and S into 2 sets one having 1 thing and second set having 3 things respectively?
205. In how many ways 4 different things can be distributed among 2 persons such that one having 3 things and other gets 1 thing respectively?
206. In how many ways 4 different things can be divided P, Q, R and S into 2 sets such that each having 2 things respectively?
207. In how many ways 4 different things can be distributed between 2 persons such that each gets 2 things respectively?

ARRANGEMENT INTO GROUPS

208. In how many different ways, 5 different balls can be arranged into 3 different boxes so that no box remains empty?
209. Five balls of different colors are to be placed in three boxes of different sizes. Each box can hold all five balls. In how many different ways can we place the balls so that no box remains empty?
210. In how many ways, 4 different balls can be distributed into 2 boxes so that no box remains empty?
211. In how many ways, 5 different balls can be distributed into 3 boxes so that no box remains empty?
212. In how many ways, 6 different balls can be distributed into 3 boxes so that no box remains empty?
213. In how many ways, can 10 apples can given to 3 students if each child should get at least one apple?
214. In how many ways, 5 identical balls can be distributed into 3 different boxes so that no box remains empty?
215. Find the number of ways of distributing 4 red balls between two persons when each person get at least one ball.
216. In how many ways, 20 apples can be distributed to 4 students in such a way that each boy can receive any number of apples?
217. A box contains 25 balls in which 10 are identical. Find the number of ways of selecting 12 balls from the box.
218. A bag contains 15 balls of which 6 are identical. Find the number of ways of selecting
 - (i) 8 balls.
 - (ii) 4 balls.
219. Find the number of ways to put 2 letters in 2 addressed envelopes so that all are in wrong envelopes.
220. Find the number of ways to put 3 letters in 3 addressed envelopes so that all are in wrong envelopes.
221. Find the number of ways to put 4 letters in 4 addressed envelopes so that all are in wrong envelopes.
222. In how many ways, we can put 5 different coloured balls into 5 boxes of colours same as that of the balls so that no ball goes to the box of same colour?
223. Let two sets A and B having $\{1, 2, 3, 4, 5\}$ elements respectively such that $f(i) \neq i$. Find the number of functions between two sets.
224. In how many ways, 5 pair of hand shocks can be distributed among 5 persons such that every person gets an odd pair?
225. There are 5 letters and 5 directed envelopes. In how many ways can all the letters be placed in which 2 are rightly placed?
226. There are 5 boxes of 5 different colours. Also there are 5 balls of colours same as those of the boxes. In how many ways, we can place 5 balls in 5 boxes such that at least 2 balls are placed in boxes of the same colour?
227. In how many ways 6 letters can be placed in 6 envelopes such that atmost 3 letters are placed in wrong envelopes?

228. Six friends go to a birthday party. They leave their coats in the lounge and pick them while returning back. In how many ways can they pick the coats so that exactly one of them picks his own coat?

MULTINOMIAL THEOREM

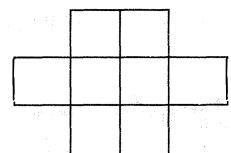
229. Find the number of selections and arrangements of 4 letters taken from the word ENGINEERING.
230. Find the total number of ways of selecting 5 letters of the word INDEPENDENT.
231. In how many ways, can three persons each throwing a single die once, make a sum of 15?
232. In how many ways, 16 identical things can be distributed among 4 persons if each person gets at least 3 things?

SOLUTIONS OF THE EQUATION WITH THE HELP OF MULTINOMIAL THEOREM

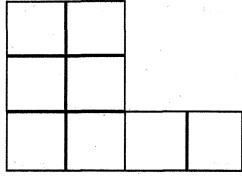
233. Find the number of solutions of $x + y + z + w = 20$ such that
 - (i) zero value are included
 - (ii) zero value are excluded.
234. Find the number of non-negative solutions of $3x + y + z = 24$
235. Find the number of non-negative integral solutions of $2x + 2y + z = 10$.
236. Find the number of non-negative solutions of the equation $x + y + z + t = 29$, where $x \geq 1, y \geq 2, z \geq 3, t \geq 4$.
237. Find the number of non-negative integral solutions to the system of equations $x + y + z + t + u = 20, x + y + z = 5$.
238. Find the number of ordered triplets of positive integers which satisfy the inequality $20 \leq x + y + z \leq 50$.
239. Find the number of points (x, y, z) in space whose each co-ordinate is a negative integer such that $x + y + z + 12 = 20$
240. Find the number of positive integral solutions of $x_1 x_2 x_3 = 30$
241. Find the number of positive integral solutions of $x_1 x_2 x_3 x_4 = 210$
242. In how many ways, can a die be thrown thrice by a person to make a sum of 12?
243. In how many ways, your teacher can distribute 10 chocolates into 4 boys if every boy can get at least one chocolate?

SELECTION OF SQUARES

244. Six Xs have to be placed in the squares of the adjoining figure, such that each row contains at least one X. In how many different ways can this be done?



245. In how many ways, the letters of the word PERSON can be placed in the squares of the adjoining figure so that no row remains empty?



GEOMETRICAL PROBLEMS

246. There are 12 points in a plane. Out of these 5 points are in a straight line and with the exception of these 4 points no other 3 points are in the same straight line. Find
- the number of straight lines formed.
 - the number of triangles formed
 - the number of quadrilaterals formed by joining these 12 points.
247. Find the number of diagonals of a polygon of decagon.
248. If the number of diagonals of a polygon is 35, find the number of sides of the polygon.
249. Find the number of triangles whose vertices are at the vertices of an octagon but none of whose sides happen to come from octagon.
250. 10 parallel lines in a plane are intersected by a family of 8 parallel lines. Find the number of parallelogram so formed.
251. Let T_n denotes the number of triangles which can be formed using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, find n .
252. Find the number of rectangles of size 7×4 .
253. Find the number of rectangles excluding squares from a rectangle of size 9×6 .
254. Find the number of points of intersection of 5 circles.
255. Find the number of points of intersection of 8 lines and 4 circles.
256. A rectangle with sides $(2n - 1)$ and $(2m - 1)$ is divided into square of unit length. Find the number of rectangles which can be formed with sides of odd length.
257. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, find the value of n .
258. The sides AB , BC and CA of a triangle ABC have 3, 4 and 5 interior points, respectively, on them. Find the number of triangles that can be constructed using these interior points.
259. Find the number of ways 2 squares can be chosen out of 8×8 chess board such that they have only one corner in common.

HINTS AND SOLUTIONS

LEVEL I

1. We have,

$$\begin{aligned} \frac{x}{5!} + \frac{x}{6!} &= \frac{1}{7!} \\ \Rightarrow \frac{x}{5!} + \frac{x}{6 \times 5!} &= \frac{1}{7 \times 6 \times 5!} \\ \Rightarrow x + \frac{x}{6} &= \frac{1}{7 \times 6} \\ \Rightarrow \frac{7}{6}x &= \frac{1}{7 \times 6} \\ \Rightarrow x &= \frac{1}{49} \end{aligned}$$

2. We have,

$$\begin{aligned} \frac{(2n)!}{n!} &= \frac{2n(2n-1)(2n-2) \dots 6.5.4.3.2.1}{n!} \\ &= \frac{\{2.4.6 \dots (2n-2)2n\} \times \{1.3.5 \dots (2n-1)\}}{n!} \\ &= \frac{2^n \{1.2.3 \dots (n-1) \cdot n\} \times \{1.3.5 \dots (2n-1)\}}{n!} \\ &= \frac{2^n \times (n)! \times \{1.3.5 \dots (2n-1)\}}{n!} \\ &= 2^n \times \{1.3.5 \dots (2n-1)\} \end{aligned}$$

3. We have,

$$\begin{aligned} \frac{3.6.9.12 \dots (3n-3)3n}{3^n} &= \frac{3^n \{1.2.3 \dots (n-1)n\}}{3^n} \\ &= \frac{3^n \times (n)!}{3^n} \\ &= (n)! \end{aligned}$$

4. Since the unit digit of a factorial more than 4 is zero, so the unit digit of the given expression

$$\begin{aligned} &= 1! + 2! + 3! + 4! + \dots + (10)! \\ &= \text{Unit digit of } 1! + 2! + 3! + 4! \\ &= \text{Unit digit of } (1 + 2 + 6 + 24) \\ &= \text{Unit digit of } 33 \\ &= 3 \end{aligned}$$

5. As we know that $5! = 120$ and $4! = 24$

Thus, e must be 4

Therefore, $N = a! + b! + c! + d! + e!$

$$= 0! + 1! + 2! + 3! + 4!$$

$$\begin{aligned} &= 1 + 1 + 2 + 6 + 24 \\ &= 34 \end{aligned}$$

6. We have $1! \times 2! = 2$

Thus, the maximum value of n is 2.

7. We have,

$$\begin{aligned} N &= 1! + 2! + 3! + 4! + 5! + 6! \\ &= 1 + 2 + 6 + 24 + 120 + 720 \\ &= 873 \end{aligned}$$

Thus, the maximum value of N is 6.

8. We have $N = a! + b! + c!$

Since the unit digit of N is 9, then

$$a = 1, b = 2 \text{ and } c = 3$$

Therefore,

$$\begin{aligned} \{a! \times b! \times c!\} &= \{1! \times 2! \times 3!\} \\ &= \{1 \times 1 \times 2 \times 1 \times 2 \times 3\} \\ &= 12 \end{aligned}$$

9. Since the last two digits of all the factorials more than 9 is 00, so the sum of all prime factorials

$$\begin{aligned} &= 2! + 3! + 5! + 7! \\ &= 2 + 6 + 120 + 5040 \\ &= 5168 \end{aligned}$$

Thus, the last two digits is 68.

10. Since $x + y + z = 25$, so the possible values of $xyz = 9, 9, 7$ and $9, 8, 8$. Hence,

the required sum

$$\begin{aligned} &= 3 \times 9! \times 9! \times 7! + 3 \times 9! \times 8! \times 8! \\ &= 3 \times 9! \times 9 \times 8! \times 7! + 3 \times 9! \times 8! \times 8 \times 7! \\ &= 3 \times 9! \times 8! \times 7!(9 + 8) \\ &= 3 \times (9 + 8) \times 9! \times 8! \times 7! \\ &= 51 \times 9! \times 8! \times 7! \end{aligned}$$

11. We have,

$$\begin{aligned} &1.1! + 2.2! + 3.3! + 4.4! \dots + n.n! \\ &= (2-1).1! + (3-1).2! + (4-1).3! + \dots \\ &\quad + (n+1).n! \\ &= (2! - 1!) + (3! - 2!) + (4! - 3!) + \dots \\ &\quad + ((n+1)! - n!) \\ &= ((n+1)! - 1!) \\ &= (n+1)! - 1 \end{aligned}$$

12. We have,

$$\begin{aligned} &\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \dots + \frac{99}{100!} \\ &= \frac{(2-1)}{2!} + \frac{(3-1)}{3!} + \frac{(4-1)}{4!} \\ &\quad + \frac{(5-1)}{5!} + \dots + \frac{(100-1)}{100!} \\ &= \left(1 - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \left(\frac{1}{3!} - \frac{1}{4!}\right) + \dots + \left(\frac{1}{99!} - \frac{1}{100!}\right) \\ &= \left(1 - \frac{1}{100!}\right) \end{aligned}$$

13. We have,

$$\begin{aligned} E_3(10!) &= \left[\frac{10}{3} \right] + \left[\frac{10}{3^2} \right] + \left[\frac{10}{3^3} \right] \\ &= 3 + 1 + 0 \\ &= 4 \end{aligned}$$

Thus, the exponent of 3 in $(10)!$ is 4.

14. We have,

$$\begin{aligned} E_5((50)!) &= \left[\frac{50}{5} \right] + \left[\frac{50}{5^2} \right] + \left[\frac{50}{5^3} \right] \\ &= 10 + 2 + 0 \\ &= 12 \end{aligned}$$

Thus, the exponent of 5 in $(50)!$ is 12.

15. We have,

$$\begin{aligned} &\text{Highest power of 10 in } 50! + 60! + 70! \\ &= \text{Highest power of 5 in } 50! + 60! + 70! \\ &= \left[\frac{50}{5} \right] + \left[\frac{50}{5^2} \right] + \left[\frac{50}{5^3} \right] \\ &= 10 + 2 + 0 \\ &= 12 \end{aligned}$$

16. Highest power of 10 in $50! \times 60! \times 70!$

$$\begin{aligned} &= \text{Highest power of 5 in } 50! \times 60! \times 70! \\ &= \text{Highest power of 5 in } 50! \\ &\quad + \text{the highest power of 5 in } 60! \\ &\quad + \text{the highest power of 5 in } 70! \\ &= \left[\frac{50}{5} \right] + \left[\frac{50}{5^2} \right] + \left[\frac{50}{5^3} \right] + \left[\frac{60}{5} \right] + \left[\frac{60}{5^2} \right] + \left[\frac{60}{5^3} \right] \\ &\quad + \left[\frac{70}{5} \right] + \left[\frac{70}{5^2} \right] + \left[\frac{70}{5^3} \right] \\ &= (10 + 2 + 0) + (12 + 2 + 0) + (14 + 2 + 0) \\ &= 42 \end{aligned}$$

17. Highest power of 10 in

$$10! + 20! + 30! + 40! + \dots + (2020)!$$

$$\begin{aligned} &= \text{Highest power of 10 in } 10! \\ &= \text{Highest power of 5 in } 10! \\ &= \text{Exponent of 5 in } 10! \end{aligned}$$

$$\begin{aligned} &= \left[\frac{10}{5} \right] + \left[\frac{10}{5^2} \right] \\ &= 2 + 0 \\ &= 2 \end{aligned}$$

18. Now,

Highest power of 2 in $33!$ = the exponent of 2 in $33!$

$$\begin{aligned} &= \left[\frac{33}{2} \right] + \left[\frac{33}{2^2} \right] + \left[\frac{33}{2^3} \right] + \left[\frac{33}{2^4} \right] + \left[\frac{33}{2^5} \right] + \left[\frac{33}{2^6} \right] \\ &= 16 + 8 + 4 + 2 + 1 + 0 \\ &= 31 \end{aligned}$$

Clearly, $33!$ is divisible by 2^{31} .

So, it is divisible by 2^{19} .

19. The number of zeroes at the end of $(100)!$

$$= \text{Highest power of 10 in } (100)!$$

= Highest power of 5 in $(100)!$

= Exponent of 5 in $(100)!$

$$\begin{aligned} &= \left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] + \left[\frac{100}{5^3} \right] \\ &= 20 + 4 + 0 \\ &= 24 \end{aligned}$$

Thus, the number of zeroes at the end of $(100)!$ is 24.

20. It is given that the highest power of 10 in $N!$ is 16.

$$\text{Thus, } \left[\frac{N}{5} \right] + \left[\frac{N}{5^2} \right] = 16$$

$$\Rightarrow 70 \leq N \leq 74$$

$$\Rightarrow N = 70, 71, 72, 73, 74.$$

When $N = 74$,

Highest power of 10 in $(74 + 1)! = 75!$

= Highest power of 5 in $75!$

= Exponent of 5 in $75!$

$$\begin{aligned} &= \left[\frac{75}{5} \right] + \left[\frac{75}{5^2} \right] + \left[\frac{75}{5^3} \right] \\ &= 15 + 3 + 0 \\ &= 18. \end{aligned}$$

21. As we know that, the highest power of 5 in $(50)!$

$$\begin{aligned} &= \left[\frac{50}{5} \right] + \left[\frac{50}{5^2} \right] + \left[\frac{50}{5^3} \right] \\ &= 10 + 2 + 0 \\ &= 12 \end{aligned}$$

So, the highest power of 5 in $(45)!$

$$\begin{aligned} &= \left[\frac{45}{5} \right] + \left[\frac{45}{5^2} \right] + \left[\frac{45}{5^3} \right] \\ &= 9 + 1 + 0 \\ &= 10 \end{aligned}$$

Thus, m should be 5.

Therefore, the minimum value of n is 45.

22. Given $N = 1.10.15 \dots 500$

$$= 5^{100} \times \{1.2.3.4 \dots 100\}$$

$$= 5^{100} \times (100)!$$

Now, Highest power of 10 in N

= Highest power of 2 in N

= Exponent of 2 in N

= Exponent of 2 in $(100)!$

$$\begin{aligned} &= \left[\frac{100}{2} \right] + \left[\frac{100}{2^2} \right] + \left[\frac{100}{2^3} \right] + \left[\frac{100}{2^4} \right] \\ &\quad + \left[\frac{100}{2^5} \right] + \left[\frac{100}{2^6} \right] + \left[\frac{100}{2^7} \right] \\ &= 50 + 25 + 12 + 6 + 3 + 1 + 0 \\ &= 97 \end{aligned}$$

Also, the exponent of 5 in $(100)!$

$$\begin{aligned} &= \left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] + \left[\frac{100}{5^3} \right] \\ &= 20 + 4 + 0 = 24 \end{aligned}$$

Thus, the highest power of 5 in $5^{100} \times (100)!$
 $= 100 + 24$
 $= 124.$

Therefore, the highest power of 10 in N is
 $= \min \{ \text{exp. of 5, exp. of 2} \}$
 $= 97$

23. There are 3 different ways you can come from your home to Sakchi and 4 different ways you can come from Sakchi to MIIT-JEE Institute.

Thus, there are $3 \times 4 = 12$ ways you can come from your home to MIIT-JEE Institute.

24. There are 5 different ways the student can enter the class room and 4 different ways, the student can leave the class room.

Thus, there are $5 \times 4 = 20$ different ways the student can enter the room and leave the room.

25. There are three places in the queue to fill one for each student.

First place can be fill 3 different ways, whereas the second and third places can be filled in 2 and 1 different ways.

Thus, there are $3 \times 2 \times 1 = 6$ different ways the queued could be formed.

26. The first prizes can be given in 5 ways, whereas second and third prizes can be given in 4 and 3 ways.

Thus, the total number of ways first three prizes can be given in $5 \times 4 \times 3 = 60$ ways.

27. A student can give answers in 5 and 7 different ways in section A and B respectively.

Thus, the total possible ways a student can give the answers $= 5 \times 7 = 35$ different ways.

28. The student first place in his shelf can be filled in 5 different ways, second place can be filled in 4 different ways, third place can be filled in 3 different ways, fourth place can be filled in 2 different ways, whereas the last place can be filled in 1 different way.

Thus, the total different ways he can arrange his books in his shelf

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 120$$

29. Clearly, there are 999 3-digit numbers lying between 100 and 1000.

The first place can be filled in 9 different ways, whereas the second and third place can be filled in 9 and 8 different ways respectively.

Therefore, the number of three digit integers
 $= 9 \times 9 \times 8$
 $= 648$

30. Therefore, the number of terms in the given product
 $= 2 \times 3 \times 4 \times 5$
 $= 120$

31. The first place can be filled in 5 different ways, second place can be filled in 4 different ways, whereas the last place can be filled in 3 different ways.

Therefore, total numbers can be made

$$= 5 \times 4 \times 3$$

$$= 120$$

32. Case I: When $x < y$

The possible solutions are (1, 9), (2, 8), (3, 7), (4, 6).

Case II: When $x > y$

The possible solutions are (9, 1), (8, 2), (7, 3), (6, 4).

Case III: When $x = y$

The possible solution is only (5, 5).

Hence, the total positive integral solutions

$$= 4 + 4 + 1$$

$$= 9$$

33. The number of ways a student can attempt a question either from section A or section B

$$= 7 + 6$$

$$= 13$$

34. A class monitor can be selected from boys in 15 different ways, whereas from girls 10 different ways.

Thus, the number of ways a class monitor can be selected either from a boy or from a girl

$$= 15 + 10 = 25$$

35. Total 3-digit number made by the digits 3 or 4 or 5

$$= 3 + 3 + 3$$

$$= 9$$

36. Number of 1-digit number = 5

Number of 2-digit numbers $= 5 \times 4 = 20$

Number of 3-digit numbers $= 5 \times 4 \times 3 = 60$

Number of 4-digit numbers $= 5 \times 4 \times 3 \times 2 = 120$

Number of 5-digit numbers $= 5 \times 4 \times 3 \times 2 \times 1 = 120$

Therefore, total positive numbers

$$= 5 + 20 + 60 + 120 + 120$$

$$= 325$$

37. Case I: When numbers are odd

Here, the unit digit can be filled in 3 different ways.

Number of odd numbers $= 3 \times (4 \times 3) = 36$

Case II: When numbers are even

Here, the unit digit can be filled in 2 different ways.

Number of even numbers $= 2 \times (4 \times 3) = 24$

Thus, the total 3-digit distinct numbers

$$= 36 + 24 = 60$$

38. Thus, the positive integral solutions $= 3! + \frac{3!}{2!} + \frac{3!}{3!}$

$$= 6 + 3 + 1$$

$$= 10$$

39. The required number of possible ways $= {}^8P_3$

$$= \frac{8!}{5!}$$

$$= 8.7.6$$

$$= 336$$

40. The odd digits are 1, 3, 5, 7, 9.

The 3-digit numbers between 100 and 999

$$= {}^5P_3 = 5.4.3 = 60$$

41. The given digits are 1, 2, 3, 4 and 5.
A number will be even, if the unit digit is either even or zero.
Thus, the digit can be filled in 2 different ways
Hence, the total even numbers $= 2 \times {}^4P_2$
 $= 2.4.3$
 $= 24$
42. The total number of possible ways 6 flags of different colours to form a signal of 4 colours $= {}^6P_4$
 $= 6.5.4.3$
 $= 360$
43. The total number of possible flags $= {}^6P_4$
 $= 6.5.4.3$
 $= 360$
44. The total number of possible ways they can take their seats $= {}^6P_6$
 $= 6!$
 $= 6.5.4.3.2.1$
 $= 720$
45. The total number of ways 3 men can stay in the hotel room $= {}^8P_3$
 $= 8.7.6$
 $= 336$
46. The total number of possible ways
 $= {}^{15}P_3$
 $= 15.14.13 = 2730$
47. The total number of arrangements of the letters of the word TABLE $= {}^5P_5$
 $= 5!$
 $= 120$
48. We have, $a = {}^4P_4 = 4! = 24$
Similarly, $b = {}^4P_4 = 4! = 24$
and $c = {}^4P_4 = 4! = 24$
Thus, $a + b + c + 10 = 24 + 24 + 24 + 10$
 $= 82$
49. The total number of 4-digit numbers
 $= 5 \times 4 \times 3 \times 2$
 $= 120$
50. The given digits are 3, 4, 5, 6 and 7.
The unit place can be filled in 3 different ways.
Hence, the total odd numbers can be made from the given digits $= 3 \times {}^4P_3$
 $= 3.4.3.2.1$
 $= 72$
51. Here, the first place can be filled in 5 different ways.
Hence, the total numbers $= 5 \times {}^5P_2$
 $= \frac{5 \times 5!}{3!}$
 $= 5 \times 20$
 $= 100$
52. Here, the first place can be filled in 4 different ways.
Thus, the total 3-digit numbers $= 4 \times {}^5P_2$
 $= 4.5.4$
 $= 80$
53. Case I: When unit digit is 0
So, the unit place can be filled in 1 way
Thus, the number of numbers $= 6 \times 5 \times 4 \times 1$
 $= 120$
Case II: When unit digit is not 0
So, the unit place can be filled in 3 different ways
and thousand's place can be filled in 5 different ways
whereas hundreds and ten's place can be filled in 5 and 4 different ways.
Thus, the number of numbers $= 5 \times 5 \times 4 \times 3$
 $= 300$
Hence, the total 4-digit numbers $= 120 + 300$
 $= 420$
54. Here, unit place can be filled in 2 different ways. Therefore, the number of even numbers $= 2 \times {}^4P_4$
 $= 2 \times 4!$
 $= 2 \times 24$
 $= 48$
55. Given digits are 1, 3, 5, 7 and 9.
The unit place can be filled in only 1 way.
Hence, the number of odd numbers divisible by 5
 $= 1 \times {}^4P_4$
 $= 1 \times 4! = 24$
56. Here, we put the unit place only zero (0).
The unit place can be filled in only 1 way.
Hence, the number of even numbers divisible by 5
 $= 1 \times {}^8P_4$
 $= \frac{8!}{4!} = 8.7.6.5 = 1680$
57. Number of 1-digit numbers $= 4$
Number of 2-digit numbers $= 4 \times {}^4P_1 = 4 \times 4 = 16$
Number of 3-digit numbers $= 4 \times {}^4P_2 = 4 \times 4 \times 3$
 $= 48$
Number of 4-digit numbers $= 4 \times {}^4P_3$
 $= 4 \times 4 \times 3 \times 2 = 96$
Number of 5-digit numbers $= 4 \times {}^4P_4$
 $= 4 \times 4! = 4 \times 24 = 96$
Hence, the total positive distinct numbers
 $= 4 + 16 + 48 + 96 + 96$
 $= 260$
58. Case I: When 2 occurs at the first place
The first place can be filled in 1 way.
Thus, the total numbers
 $= 1 \times {}^4P_3 = 1 \times 4! = 24$
Case II: When 3, 4, 5 and 6 occur at the first place
The first place can be filled in 4 different ways.
Thus, the total numbers $= 4 \times {}^4P_3 = 4 \times 4! = 96$
Hence, the total number of numbers $= 24 + 96$
 $= 120$

59. The number of one one functions from A to $B = {}^7P_4$

$$= \frac{7!}{3!} = 7 \times 5!$$

$$= 840$$

60. Hence, total number of one one onto functions

$$= {}^5P_5 = 5! = 120$$

61. As we know that, the inverse of a function exists only when that function is one one and onto.

So, the number of inverse functions

= the number of one one onto functions

$$= {}^4P_4 = 4! = 24$$

62. The number of functions from A to $B = 3^5$

So, the number of into functions from A to B

$$= 2^5 + 2^5 + 2^5 - 3$$

$$= 32 + 32 + 32 - 3$$

$$= 96 - 3$$

$$= 93$$

The number of onto functions from A to B

= Total functions – into functions

$$= 3^5 - 93$$

$$= 243 - 93$$

$$= 150$$

63. Number of onto functions between two sets = Number of distribution of 5 balls into 4 boxes where no ball remains empty.

Here, we can make a quadruplet on 5 is $5 \rightarrow (1, 1, 1, 4)$

Thus, the number of onto functions

$$= \frac{5!}{1!1!1!2!} \times \frac{4!}{3!}$$

$$= \frac{5!}{2!} \times \frac{4!}{3!}$$

$$= 60 \times 4$$

$$= 240$$

64. The sum of the digits in the unit place

$$= (4-1)! \times (2+3+4+5)$$

$$= 6 \times 14 = 84$$

65. The required sum

$$= (4-1)! \times (1+2+3+4) \times \left(\frac{10^4-1}{9} \right)$$

$$= 6 \times 10 \times 1111$$

$$= 66660$$

66. The required sum

$$= (4-1)! \times (0+1+2+3) \times \left(\frac{10^4-1}{9} \right)$$

$$- (3-1)! \times (1+2+3) \times \left(\frac{10^3-1}{9} \right)$$

$$= 6 \times 6 \times 1111 - 2 \times 6 \times 111$$

$$= 36 \times 1111 - 12 \times 111$$

$$= 39996 - 1332$$

$$= 38664$$

67. Hence, the required sum is

$$= \frac{(4-1)!}{2} \times (2+3+3+4) \times \left(\frac{10^4-1}{9} \right)$$

$$= 3 \times 12 \times 1111$$

$$= 36 \times 1111$$

$$= 39996$$

68. The total number of possible ways

$$= 2 \times 2 \times 2 \times \dots \times 2 \text{ (n times)}$$

$$= 2^n$$

69. The first delegate can stay in any one of 5 hotels. So, it can be possible in 5 different ways.

So, the 2nd, 3rd and 4th delegates have also 5 choices.

Hence, the total number of possible ways

$$= 5 \times 5 \times 5 \times 5$$

$$= 5^4 = 625$$

70. (i) The total number of ways, 3 friends can stay in a 5 hotel of a two n

$$= 5 \times 5 \times 5 \times 5^3 = 125$$

- (ii) The total number of ways, when no two friends stay in a same hotel

$$= 5 \times 4 \times 3 = 60$$

71. The total possible ways, 5 rings can be worn on the four fingers of one hand

$$= 5 \times 5 \times 5 \times 5 = 5^4 = 625$$

72. Here, m = the number of ways a child put 3 marbles in his 4 pockets

$$= 4 \times 4 \times 4 = 64$$

and n = the number of 3-digit distinct number

$$= 5 \times 4 \times 3 = 60$$

Therefore

$$m + n + 10 = 64 + 60 + 10$$

$$= 134$$

73. The total number of possible ways

$$= 10 \times 10 \times 10 \times 10 \times 10 = 10^5$$

74. The total number of possible ways

$$= 9 \times 10 \times 10 \times 10$$

$$= 9 \times 10^3$$

$$= 9000$$

75. The 3 letters code words can be made from the word INTEx

$$= 5 \times 5 \times 5 = 125$$

76. The total number of possible ways

$$= 4 \times 4 \dots \times 4 \text{ (10 times)}$$

$$= 4^{10}$$

77. Given that, each question has four choices a, b, c and d respectively.

Here, the option a is either correct or incorrect.

So, there are 2 ways, the student can give the answer for the option a .

Similarly, for the options b, c and d .

Thus, a particular questions have $2 \times 2 \times 2 \times 2 = 16$ possible answers.

It will be considered only when all the options are incorrect.

Thus, the total number of possible answers = 15.

Since the total number of questions is 10, so the required number of possible answers

$$= 15 \times 15 \times 15 \times \cdots \times 15 \text{ (10 times)}$$

$$= 15^{10}$$

78. The required number of permutations of the letters of

$$\text{the word} = \frac{3!}{2!} = \frac{6}{2} = 3$$

79. The required number of permutations of the letters of

$$\text{the word} = \frac{5!}{3!} = \frac{120}{6} = 20$$

80. There are 2 Ms, 2 As, 2 Ts and 1 H, 1 I, 1 C, 1 S and 1 E.

Hence, the required permutations of the letters of the word

$$= \frac{11!}{2! \times 2! \times 2!}$$

81. There are 2 Os, 3 Ts, 2 Ns, 2 Is and 1 C, 1 S and 1 U.

Hence, the required permutations of the letters of the

$$\text{word} = \frac{12!}{2! \times 3! \times 2! \times 2!}$$

82. The required number of seven-digit numbers

$$= \frac{7!}{3! \times 2!}$$

83. The required sum

$$= \frac{(6-1)!}{2! \times 3!} \times (2+3+3+4+4+4) \times \left(\frac{10^6-1}{9} \right)$$

$$= \frac{120}{12} \times 20 \times 111111$$

$$= 22222200$$

84. We have, $m = \frac{7!}{2!} = \frac{5040}{2} = 2520$

$$\text{and } n = \frac{3!}{2!} = \frac{6}{2} = 3$$

Hence, the value of

$$\begin{aligned} m+n+10 &= 2520+3+10 \\ &= 2533 \end{aligned}$$

85. We have 10 students can be arranged themselves in 10! ways.

Out of them 1/2 ways A is ahead of B and another 1/2 ways B is ahead of A.

$$\begin{aligned} \text{Hence, the required number of ways} &= \frac{10!}{2} \\ &= 1814400 \end{aligned}$$

86. 10 persons can be arranged in 10! ways.

Out of them 1/2 ways A before B, 1/2 ways B before C and also out of them 1/2 ways C before D.

Hence, the required number of possible ways

$$\begin{aligned} &= \frac{10!}{8} \\ &= 453600 \end{aligned}$$

87. Number of permutations of $a, b, c, d = 4! = 24$

Number of permutations of $ab, c, d = 3! = 6$

Number of permutations of $a, bc, d = 3! = 6$

Number of permutations of $a, b, cd = 3! = 6$

Number of permutations of b and $c = 2! = 2$

Number of permutations of c and $d = 2! = 2$

Number of permutations of d and $b = 2! = 2$

Number of permutations of b, c and $d = 1$

Hence, the required number of permutations

$$\begin{aligned} &= 4! - 3! - 3! - 3! + 2! + 2! + 2! - 1! \\ &= 11 \end{aligned}$$

88. Here, 2 boys can be tied by a string and consider thing

Total number of things = $(10 - 2 + 1) = 9$

Hence, the total number of ways they can sit = $9! \times$

89. Here, 6 boys can be tied by a string and consider thing. In a similar way 5 girls are tied by a string : also consider as 1 thing.

Thus, total number of things = 2

Hence, the total number of ways boys and girls can together = $6! \times 5! \times 2!$

90. There are 3 vowels (A, I, O) and 5 consonants (F, R, T, N) in the given word.

3 vowels can be tied by a string and consider 1 thing

Total number of things = $5 + 1 = 6$

Hence, the number of ways, it can be arranged = $6! \times$

91. There are 5 vowels (A, E, I, O, U) and 3 consonants (Q, T, N) in the given word.

5 vowels can be tied by a string and consider 1 thing

Total number of things = $3 + 1 = 4$

Hence, the number of ways, it can be arranged = $4! \times$

92. There are 2 vowels (A, U) and 3 consonants (L, G, in the given word.

2 vowels can be tied by a string and consider 1 thing similarly 3 consonants will consider as 1.

Total number of things = $1 + 1 = 2$

Hence, the number of ways, it can be arranged

$$\begin{aligned} &= 2 \times 2! \times 3! \\ &= 24 \end{aligned}$$

93. There are 4 vowels (U, I, E, I) and 6 consonants (N, R, S, T, Y).

When 4 vowels are together, consider the 4 vowels as thing.

Total number of things = $6 + 1 = 7$

Hence, the number of ways, it can arranged

$$\begin{aligned} &= 7! \times \frac{4!}{2!} \\ &= 7! \times 12 \\ &= 12 \times 7! \end{aligned}$$

94. A, B and C can be arranged themselves in $3!$ ways in which C is ahead of A and B in 2 ways
Hence, the number of possible arrangements

$$\begin{aligned} &= (8)! \times \frac{2}{3!} \\ &= 40320 \times \frac{1}{3} \\ &= 13440 \end{aligned}$$

95. When all vowels come together, the possible arrangements = $\frac{5! \times 3!}{2}$

The number of arrangements of the word ALGEBRA, without any restriction = $\frac{7!}{2}$

Hence, the number of ways, all vowels do not come together = Total number of possible arrangements without any restrictions – number of possible arrangements when all vowels come together

$$\begin{aligned} &= \frac{7!}{2} - \frac{5! \times 3!}{2} \\ &= 2520 - 360 \\ &= 2160 \end{aligned}$$

96. When I and N are together, total number of things = $5 + 1 = 6$

So, the possible arrangements = $\frac{6!}{2} = 380$

The number of arrangements of the word INTEGER, without any restriction = $\frac{7!}{2}$

Hence, the number of ways, I and N are never come together

$$\begin{aligned} &= \text{Total arrangements without restrictions} - \text{together} \\ &= \frac{7!}{2} - \frac{6!}{2} \\ &= 2520 - 380 \\ &= 2140 \end{aligned}$$

97. The number of arrangements of the word SUCCESS, without any restriction

$$\begin{aligned} &= \frac{7!}{2! \times 3!} \\ &= 420 \end{aligned}$$

When all S s come together, the possible arrangements

$$= \frac{5!}{2!} \times \frac{3!}{3!} = \frac{5!}{2!} = 60$$

Hence, the number of ways, all S s do not come together

$$\begin{aligned} &= \text{Total} - \text{together} \\ &= 420 - 60 \\ &= 360 \end{aligned}$$

98. When two N s come together, the possible arrangements are

$$= \frac{5!}{3!} \times \frac{2!}{2!} = \frac{5!}{3!} = 20$$

The number of arrangements of the word BANANA, without any restriction

$$= \frac{6!}{3! \times 2!} = \frac{120}{2} = 60$$

Hence, the number of ways, all N s do not appear adjacently

$$\begin{aligned} &= \text{Total} - \text{together} \\ &= 60 - 20 \\ &= 40 \end{aligned}$$

99. When both A are together, consider them as 1 unit, so

the possible arrangements are = $\frac{7!}{3!}$

When both A and all three L s are together, consider them as two separate unit.

So, the possible arrangements

$$= 5! \times \frac{2!}{2!} \times \frac{3!}{3!} = 5! = 120$$

Hence, the number of ways all L s do not come together but all A s come together.

$$= \frac{7!}{3!} - 5!$$

$$\begin{aligned} &= 7 \times 5! - 5! \\ &= (7 - 1) \times 5! \\ &= 6 \times 120 \\ &= 720 \end{aligned}$$

100. There are 2 I s, 3 E s, 2 T s, 1 N , 1 R , 1 M , 1 A and 1 D in the given word.

So, there are 6 vowels and 6 consonants.

As per the given conditions, all the consonants are together.

Consider all the consonants as 1 thing.

Total number of things = $6 + 1 = 7$ things

Hence, the number of ways it can be done

$$\begin{aligned} &= \frac{7!}{2! \times 3!} \times \frac{6!}{2!} \\ &= \frac{7!}{2!} \times \frac{5!}{2!} \end{aligned}$$

101. There are 3 N s, 3 E s, 1 I and 1 T in the given word.

Case I: When there is no restriction, the number of arrangements

$$= \frac{8!}{3! \times 3!} = 1120$$

Case II: When all vowels are together, the number of arrangements

$$= \frac{5!}{3!} \times \frac{4!}{3!} = 80$$

Case III: When all consonants are together, the number of arrangements

$$= \frac{5!}{3!} \times \frac{4!}{3!} = 80$$

Case IV: When vowels and consonants are together, the number of arrangements

$$= 2! \times \frac{4!}{3!} \times \frac{4!}{3!} = 32$$

Hence, the number of ways it can be done

$$= 1120 - (80 + 80 - 32) \\ = 1120 - 128 \\ = 992$$

102. The alphabetical order of the word TOUGH is G, H, O, T, U.

The number of words beginning with G is 4!

The number of words beginning with H is 4!

The number of words beginning with O is 4!

The number of words beginning with TG is 3!

The number of words beginning with TH is 3!

The number of words beginning with TOG is 2!

The number of words beginning with TOH is 2!

The number of words beginning with TOU is TOUGH

Hence, the rank of the word TOUGH is

$$= 4! + 4! + 4! + 3! + 3! + 2! + 1! = 1 \\ = 72 + 12 + 5 \\ = 89$$

103. The alphabetical order of IIT is I, I, T

So, the rank of the word IIT is 1.

104. The alphabetical order of AIEEE is A, E, E, E, I.

The number of words beginning with AE is $\frac{3!}{2!} = 3$

The number of words beginning with AI is AIEEE.

Hence, the rank of the word AIEEE is $= 3 + 1 = 4$

105. The alphabetic order of the word ANNA is A, A, N, N, N.

The number of words beginning with AA is 1

The number of words beginning with AN is 1

The number of words beginning with ANN is ANNA

Hence, the rank of the word ANNA $= 1 + 1 + 1 = 3$

106. The alphabetic order of the word PATNA is A, A, N, P, T.

The number of words beginning with A is 4!

The number of words beginning with N is $\frac{4!}{2!}$

The number of words beginning with PAA is 2!

The number of words beginning with PAN is 2!

The number of words beginning with PATA is PATAN

The number of words beginning with PATN is PATNA

Hence, the rank of the word PATNA

$$= 4! + \frac{4!}{2!} + 2! + 2! + 1 + 1 \\ = 42$$

107. The alphabetic order of the word SURITI is I, I, R, S, T, U.

The number of words beginning with I is 5!

The number of words beginning with R is $\frac{5!}{2!}$

The number of words beginning with SI is 4!

The number of words beginning with SR is $\frac{4!}{2!}$

The number of words beginning with ST is $\frac{4!}{2!}$

The number of words beginning with SUI is 3!

The number of words beginning with SURI is 1

The number of words beginning with SURIT is SURITI

Hence, the rank of the word SURITI is

$$= 5! + \frac{5!}{2!} + 4! + \frac{4!}{2!} + \frac{4!}{2!} + 3! + 1 + 1 \\ = 120 + 60 + 24 + 12 + 12 + 6 + 1 + 1 \\ = 180 + 56 \\ = 236$$

108. The alphabetic order of the word SANIA is A, A, I, N, S.

The number of words beginning with A is 4!

The number of words beginning with I is $\frac{4!}{2!}$

The number of words beginning with N is $\frac{4!}{2!}$

The number of words beginning with SAA is 2!

The number of words beginning with SAI is 2!

The number of words beginning with SANA is SANAI

The number of words beginning with SANI is SANIA

Hence, the rank of the word SANIA

$$= 4! + \frac{4!}{2!} + \frac{4!}{2!} + 2! + 2! + 1 + 1 \\ = 48 + 6 \\ = 54$$

109. There are 3 vowels (A, I, O) and 5 consonants (F, R, C, T, N)

Thus, there are 6 gaps in between 5 consonants

Here, we place 3 vowels in between 6 gaps in 6P_3 ways, whereas 5 consonants can be arranged themselves in 5! ways.

Thus, the total number of ways it can be done

$$= {}^6P_3 \times 5! \\ = \frac{6!}{3!} \times 5! \\ = 5! \times 5! \\ = 120 \times 120 \\ = 14400$$

110. There are 3 vowels (A, I, E) and 5 consonants (T, R, N, G, L).

So, we place 3 vowels in between 6 gaps of 5 consonants.

Hence, the number of ways it can be done

$$= {}^6P_3 \times 5! \\ = 120 \times 120 \\ = 14400.$$

111. There are 3 vowels (A, E, E) and 3 consonants (M, C, T)

So, we place 3 vowels (2 are alike) in between of 4 gaps of 3 consonants.

Hence, the number of ways it can be done

$$= \frac{{}^4P_3}{2!} \times 3! \\ = \frac{4!}{2!} \times 3! \\ = 12 \times 6 = 72$$

112. There are 3 Ns, 3 Es, 1 I and 1 T in the given word NINETEEN.

So, we place 3 Ns in between of 6 gaps of 3 Es, 1 I and 1 T.

Hence, the number of ways, it can be done

$$= \frac{{}^6P_3}{3!} \times \frac{5!}{3!} \times \frac{3!}{3!} \\ = \frac{6!}{3! \times 3!} \times \frac{5!}{3!} \\ = \frac{5!}{3!} \times \frac{5!}{3!} \\ = 20 \times 20 \\ = 400$$

113. There are 3 Ns, 3 Es, 1 I and 1 T in the given word NINETEEN.

So, there are 4 vowels and 4 consonants.

Case I: When word starts with vowel

Then vowels occupy the odd places whereas consonants take even places.

Hence, the number of ways it can be done

$$= \frac{4!}{3!} \times \frac{4!}{3!} \\ = 4 \times 4 \\ = 16$$

Case II: When word starts with consonants

Then consonants occupy the odd places, whereas the vowels take even places.

Hence, the number of ways, also it can be done

$$= \frac{4!}{3!} \times \frac{4!}{3!} \\ = 4 \times 4 \\ = 16$$

Thus, the total number of ways it can be done

$$= 16 + 16 \\ = 32$$

114. The arrangements will be of the form

B G B G B G B G B

6 boys can be arranged themselves in $6!$ ways whereas

5 girls can be arranged them in $5!$ ways.

Hence, the number of ways it can be done

$$= 6! \times 5!$$

115. Case I: When first place occupies by a boy

The seating arrangement can be of the form

B G B G B G B G B

Hence, it can be done

$$= 5! \times 5!$$

Case II: When the first place occupies by a girls

The seating arrangement can be of the form

G B G B G B G B G

Hence, it can be done

$$= 5! \times 5!$$

Thus, the total number of ways, it can be done

$$= 5! \times 5! + 5! \times 5! \\ = 2 \times 5! \times 5! \\ = 2 \times 120 \times 120 \\ = 28800$$

116. Odd digits are 3, 3, 5, 5 and even digits are

2, 2, 8, 8, 8.

Odd digits take places 2nd, 4th, 6th and 8th positions whereas even digits take places 1st, 3rd, 5th, 7th and 9th positions.

Hence, the number of nine-digit numbers can be formed from the given number

$$= \frac{4!}{2! \times 2!} \times \frac{5!}{2! \times 3!} \\ = 6 \times 10 \\ = 60$$

117. There are 2 Ms, 2 As, 2 Ts, 1 H, 1 E, 1 I, 1 C and 1 S in the given word.

So, there are 4 vowels and 7 consonants.

Hence, the number of possible arrangements

$$= \frac{4!}{2!} \times \frac{7!}{2! \times 2!} \\ = 3 \times 5040 \\ = 15120$$

118. There are 3 vowels (A, I, E) and 5 consonants (C, C, D, N, T) in the given word.

Now, 3 vowels can be arranged in $3!$ ways

So, there is only one way where they will be in given order.

$$\text{Hence, the total number of such words} = \frac{8!}{2!} \times \frac{1}{3!} \\ = 3360$$

119. There are 5 vowels (I, E, E, I, E) and 6 consonants (N, F, F, C, T, V).

Since the position of each vowel is fixed, then 6 consonants can be arranged themselves in $\frac{6!}{2!}$ ways

$$= \frac{720}{2} = 360$$

120. (i) Total number of persons = $6 + 5 = 11$
 Thus, 11 persons can be seated in a round table in $(11 - 1)! = (10)!$ ways
 (ii) Consider 5 Englishmen as a one person.
 Total number of persons = $6 + 1 = 7$
 Thus, the total number of ways, they can be seated in a round table
 $= (7 - 1)! \times 5! = 6! \times 5!$
 (iii) when 5 Englishmen do not sit together, the possible seating arrangements
 $= \text{Total} - \text{together}$
 $= (10)! - 6! \times 5!$
 (iv) 6 Indians can be seated around a round table in 5! ways.
 If no two Englishmen do not sit together, we can fill them in between 6 Indians in 6P_5 ways.
 Hence, the total number of possible seating arrangements = ${}^6P_5 \times 5!$
 $= 6! \times 5!$

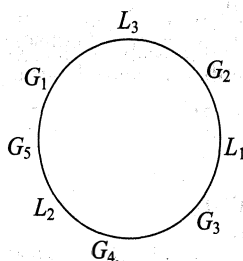
121. The number of circular arrangements of 6 different flowers = $(6 - 1)! = 5! = 120$

122. As we know that, to form a necklace, clockwise and anti-clockwise arrangements are the same.
 Hence, the required number of arrangements

$$= \frac{1}{2} \times (50 - 1)!$$

$$= \frac{49!}{2}$$

123. The five gentlemen can arrange themselves around a round table in $4!$ ways.



If every gentlemen is to have at least one lady by his side, then two ladies must be in two of the adjacent gaps and the third lady must occupy only one gap.

Thus, for every arrangement of gentlemen, ladies can be arranged themselves in $5 \times 3!$ ways = 30 ways (where two adjacent gaps may be chosen in 5 ways and the 3 ladies each time may be arranged in $3!$ ways).

Hence, the required number of ways it can be done

$$= 5 \times 3! \times 4!$$

$$= 5 \times 6 \times 24$$

$$= 720$$

124. If two persons be A and B , the remaining 18 persons would be $P_1, P_2, P_3, \dots, P_{18}$.

Take one of 18 persons, say P_1 , and make him sit between A and B , i.e. A, P_1, B .

In the similar way, we can place $P_2, P_3, \dots, P_{18}$ in between A and B .

Thus, there are 18 ways of choosing one person between A and B .

Hence, the required number of possible ways

$$= 18 \times 17! \times 2!$$

125. Here, total number of persons = $7 + 8 = 15$

Hence, the number of ways they can be seated around a circular table = $(15 - 1)! = 14!$ ways.

126. Since each of the seat is given by the number and all are different numbers, so the arrangement of this seating is a linear arrangement.

Hence, the number of ways, this seating arrangement can be done = $15!$

127. Three students A, B and C can be arranged themselves in $3!$ ways in which in 2 cases B is in between A and C .

So, total number of things = $7 + 1 = 8$

Hence, the number of ways it can be done

$$= 7! \times \frac{2}{3!}$$

$$= 7 \times 5! \times 2$$

$$= 14 \times 120$$

$$= 1680$$

128. First, we will select 10 persons out of 20 persons.

It can be done in ${}^{20}C_{10}$ ways.

Around each table 10 persons can be seated in $9!$ ways.

Hence, the number of ways they can be seated

$$= {}^{20}C_{10} \times 9! \times 9!$$

129. The number of ways 20 persons can be seated around a round table

$$= \frac{{}^{20}P_{10}}{20}$$

130. The number of ways it can be done

$$= \frac{(10)!}{2} \times 9!$$

$$= \frac{10 \times 9!}{2} \times 9!$$

$$= 5 \times (9!)^2$$

131. The number of ways it can be done

$$= \frac{1}{2} \times \frac{{}^{30}P_{20}}{20}$$

132. The number of ways it can be distributed between two students

$$= {}^7C_4 \times {}^4C_3$$

$$\begin{aligned}
 &= \frac{7!}{4! \times 3!} \times \frac{4!}{3! \times 1!} \\
 &= \frac{7.6.5.4}{6} \\
 &= 140
 \end{aligned}$$

133. The number of ways Master Rohan can invite his friends

$$\begin{aligned}
 &= {}^{10}C_4 \\
 &= \frac{10!}{4! \times 6!} \\
 &= \frac{10.9.8.7}{24} \\
 &= 210
 \end{aligned}$$

134. The number of ways I can select the students for the Olympiad examination

$$\begin{aligned}
 &= {}^{12}C_5 \\
 &= \frac{12!}{5! \times 7!} \\
 &= \frac{12.11.10.9.8}{120} \\
 &= 792
 \end{aligned}$$

135. The number of ways a team of 3 boys and 4 girls are selected

$$\begin{aligned}
 &= {}^7C_3 \times {}^9C_4 \\
 &= \frac{7!}{3!4!} \times \frac{9!}{4!5!} \\
 &= \frac{7.6.5}{6} \times \frac{9.8.7.6}{24} \\
 &= 35 \times 126 \\
 &= 4410
 \end{aligned}$$

136. The number of total selections

$$\begin{aligned}
 &= {}^5C_3 \times {}^2C_5 \\
 &= {}^5C_5 \times {}^{21}C_5 \\
 &= {}^{21}C_5
 \end{aligned}$$

137. In general, we buy a ticket for journey of two stoppages (one place to another place).

Consider the number of stoppages is n .

Thus, ${}^nC_2 = 190$

$$\Rightarrow \frac{n(n-1)}{2} = 190$$

$$\Rightarrow n(n-1) = 380$$

$$\Rightarrow n^2 - n - 380 = 0$$

$$\Rightarrow (n-20)(n+19) = 0$$

$$\Rightarrow n = 20, -19$$

Hence, the number of stoppages is 20.

138. Let n be the number of students.

Number of pairs of the students = nC_2

For nC_2 pairs, number of cards sent = 2

Thus, $2 \times {}^nC_2 = 600$

$$\Rightarrow {}^nC_2 = 300$$

$$\Rightarrow \frac{n(n-1)}{2} = 300$$

$$\Rightarrow n^2 - n - 600 = 0$$

$$\Rightarrow (n-24)(n+25) = 0$$

$$\Rightarrow n = 24, -25$$

Hence, the number of students = 24.

139. The number of times he has to go to the circus with his children

$$= {}^8C_3$$

$$= \frac{8!}{3!5!}$$

$$= \frac{8.7.6}{6}$$

$$= 56$$

140. The total number of ways they can be allotted to different wards

$$\begin{aligned}
 &= {}^{20}C_4 \times {}^{16}C_5 \times {}^{11}C_8 \\
 &= \frac{20!}{4!16!} \times \frac{16!}{5!11!} \times \frac{11!}{8!3!} \\
 &= \frac{20!}{4!5!3!8!}
 \end{aligned}$$

141. In a pack of 52 cards, there are 4 suits, i.e. spades, clubs, diamond and hearts.

Hence, the number of possible ways of selecting 4 cards from a pack of 52 cards

$$\begin{aligned}
 &= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 \\
 &= 4 \times ({}^{13}C_4)
 \end{aligned}$$

142. In a pack of 52 cards, there are 4 suits, i.e. spades, clubs, diamond and hearts.

Hence, the number of possible ways of selecting 4 cards from a pack of 52 cards

$$\begin{aligned}
 &= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 \\
 &= ({}^{13}C_4)^4 \\
 &= (13)^4
 \end{aligned}$$

143. Total number of persons = 6 + 4 = 10.

Hence, the number of ways it can be done

$$\begin{aligned}
 &= {}^{10}C_5 - {}^6C_5 \\
 &= \frac{10!}{5!5!} - \frac{6!}{5!1!} \\
 &= \frac{10.9.8.7.6}{120} - 6
 \end{aligned}$$

$$= 252 - 6$$

$$= 246$$

144. As we know that nC_r is defined only when $r \in n, r \in W, n \in N$.

$$\text{Thus, } 2x - 5 \leq x - 2, x - 2 \geq 0, 2x - 5 > 0$$

$$\Rightarrow x \leq 3, x \geq 2, x > \frac{5}{2}$$

$$\Rightarrow x = 3$$

Hence, the domain of the function $f(x) = D_f = \{3\}$.

145. As we know that, nC_r is defined only when $r \leq n, r \in W, n \in N$.

$$\text{Thus, } 3x - 7 \leq x - 1, 3x - 7 > 0, x - 1 \geq 0$$

$$\Rightarrow 3x - x \leq 7 - 1, x > \frac{7}{3}, x \geq 1$$

$$\Rightarrow 2x \leq 6, x > \frac{7}{3}, x \geq 1$$

$$\Rightarrow x \leq 3, x > \frac{7}{3}, x \geq 1$$

$$\Rightarrow x = 3$$

Thus, the domain of the function $f(x) = D_f = \{3\}$

Hence, the range of a function is ${}^2C_2 = 1$

146. We have,

$${}^{10}C_r = {}^{10}C_{r+2}$$

$$\Rightarrow r + r + 2 = 10$$

$$\Rightarrow 2r + 2 = 10$$

$$\Rightarrow 2r = 8$$

$$\Rightarrow r = 4$$

Hence, the value of r is 4.

147. Given

$${}^nC_{n-4} = 15$$

$$\Rightarrow {}^nC_4 = 15$$

$$\Rightarrow \frac{n!}{4!(n-4)!} = 15$$

$$\Rightarrow \frac{n(n-1)(n-2)(n-3)}{24} = 15$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 15 \times 24$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 360$$

$$\Rightarrow n(n-1)(n-2)(n-3) = 6 \times 5 \times 4 \times 3$$

$$\Rightarrow n = 6$$

148. We have,

$${}^{15}C_r : {}^{15}C_{r-1} = 11 : 5$$

$$\Rightarrow \frac{{}^{15}C_r}{{}^{15}C_{r-1}} = \frac{11}{5}$$

$$\Rightarrow \left(\frac{15-r+1}{r} \right) = \frac{11}{5}$$

$$\Rightarrow 11r = 75 - 5r + 5$$

$$\Rightarrow 16r = 80$$

$$\Rightarrow r = 5$$

Hence, the value of r is 5.

149. We have,

$$\begin{aligned} & {}^{20}C_{13} + {}^{20}C_{15} + {}^{20}C_9 - {}^{20}C_7 - {}^{20}C_5 - {}^{20}C_{11} \\ &= {}^{20}C_{20-13} + {}^{20}C_{20-15} + {}^{20}C_9 - {}^{20}C_7 - {}^{20}C_5 - {}^{20}C_{11} \\ &= {}^{20}C_7 + {}^{20}C_5 + {}^{20}C_{11} - {}^{20}C_7 - {}^{20}C_5 - {}^{20}C_{11} \\ &= {}^{20}C_7 + {}^{20}C_5 + {}^{20}C_{11} - {}^{20}C_7 - {}^{20}C_5 - {}^{20}C_{11} \\ &= 0 \end{aligned}$$

150. We have,

$${}^{n-1}C_3 + {}^{n-1}C_4 > {}^nC_3$$

$$\Rightarrow {}^{n-1}C_4 + {}^{n-1}C_3 > {}^nC_3$$

$$\Rightarrow {}^{n-1}C_4 + {}^{n-1}C_{4-1} > {}^nC_3$$

$$\Rightarrow {}^nC_4 + {}^nC_3$$

$$\Rightarrow n > 4 + 3$$

$$\Rightarrow n > 7$$

151. We have,

$$\begin{aligned} & {}^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3 \\ &= {}^{47}C_4 + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3) \\ &= ({}^{47}C_4 + {}^{47}C_3) + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3) \\ &= ({}^{48}C_4 + {}^{48}C_3) + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3) \\ &= ({}^{49}C_4 + {}^{49}C_3) + ({}^{51}C_3 + {}^{50}C_3) \\ &= ({}^{50}C_3 + {}^{50}C_3) + ({}^{51}C_3) \\ &= ({}^{51}C_3 + {}^{51}C_3) \\ &= {}^{52}C_4 \end{aligned}$$

152. We have,

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{84}{36} \text{ and } \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{126}{84}$$

$$\Rightarrow \frac{n-r+1}{r} = \frac{7}{3} \text{ and } \frac{n-r}{r+1} = \frac{3}{2}$$

$$\Rightarrow 10r - 3n = 3 \text{ and } 10r - 4n = -6$$

Solving, we get,

$$n = 9 \text{ and } r = 3.$$

153. We have, ${}^{x+3}P_3 = {}^{x+2}C_3 + 20$

$$\Rightarrow \frac{(x+3)!}{(x+3-2)!} = \frac{(x+2)!}{3!(x+2-3)!} + 20$$

$$\Rightarrow \frac{(x+3)!}{(x+1)!} = \frac{(x+2)!}{3!(x-1)!} + 20$$

$$\Rightarrow (x+3)(x+2) = \frac{x(x+2)(x+1)}{6} + 20$$

$$\Rightarrow 6(x+3)(x-2) = x(x+2)(x+1) + 10$$

$$\Rightarrow 6(x^2 + 5x + 6) = x(x^2 + 3x + 2) + 120$$

$$\Rightarrow x^3 + 3x^2 - 6x^2 + 2x - 30x + 120 - 36 = 0$$

$$\Rightarrow x^3 - 3x^2 - 28x + 84 = 0$$

$$\Rightarrow x^2(x-3) - 28(x-3) = 0$$

$$\Rightarrow (x-3)(x^2-28)=0$$

$$\Rightarrow x=3, \pm\sqrt{28}$$

Thus $x=3$.

154. Given,

$$m = {}^nC_2$$

$$\Rightarrow m = \frac{n(n-1)}{2}$$

$$\text{Thus, } {}^nC_2 = \frac{m(m-1)}{2}$$

$$= \frac{\left(\frac{n^2-n}{2}\right)\left(\frac{n^2-n}{2}-1\right)}{2}$$

$$= \frac{(n^2-n)(n^2-n-2)}{8}$$

$$= \frac{n(n-1)(n-2)(n+1)}{8}$$

155. We have,

$$\frac{1}{{}^4C_n} = \frac{1}{{}^5C_n} + \frac{1}{{}^6C_n}$$

$$\Rightarrow \frac{n!(4-n)!}{4!} = \frac{n!(5-n)!}{5!} + \frac{n!(6-n)!}{6!}$$

$$\Rightarrow \frac{(4-n)!}{4!} = \frac{(5-n)!}{5!} + \frac{(6-n)!}{6!}$$

$$\Rightarrow \frac{(5-n)!}{5!} + \frac{(6-n)!}{6!} = \frac{(4-n)!}{4!}$$

$$\Rightarrow \frac{(5-n)}{5} + \frac{(6-n)(5-n)}{6 \times 5} = 1$$

$$\Rightarrow 6(5-n) + (6-n)(5-n) = 30$$

$$\Rightarrow (5-n)(12-n) = 30$$

$$\Rightarrow (n-5)(n-12) = 30$$

$$\Rightarrow n^2 - 17n + 60 - 30 = 0$$

$$\Rightarrow n^2 - 17n + 30 = 0$$

$$\Rightarrow (n-2)(n-15) = 0$$

$$\Rightarrow n = 2, 15$$

Hence, the value of n is 2.

156. (i) Since two particular students are always chosen, the selection is needed for another 18 students from 98 students.

Hence, the required number of selections = ${}^{98}C_{18}$

(ii) Since five particular students are never chosen, the selection is needed for 20 students from 95 students.

Hence, the required number of selection is = ${}^{95}C_{20}$

157. The required number of possible selection

$$= {}^4C_3 \times {}^5C_4 + {}^4C_4 \times {}^5C_4 + {}^4C_3 \times {}^5C_5 + {}^4C_4 \times {}^5C_5$$

$$= 4 \times 5 + 1 \times 5 + 4 \times 1 + 1 \times 1$$

$$= 20 + 5 + 4 + 1$$

$$= 30$$

158. As we know that, a pack of cards consists of 52 cards in which 4 are aces and 4 are kings.

Hence, the required number of possible selections

$$= {}^4C_1 \times {}^4C_1 \times {}^4C_{44}$$

$$= 4 \times 4 \times \frac{44!}{4! \times 40!}$$

$$= 4 \times 4 \times \frac{44 \times 43 \times 42 \times 41}{4 \times 3 \times 2 \times 1}$$

$$= 44 \times 43 \times 14 \times 41$$

159. The required number of possible selection

$$= {}^4C_3 \times {}^{48}C_9 + {}^4C_2 \times {}^{48}C_8 + {}^4C_3 \times {}^{48}C_7 + {}^4C_4 \times {}^{48}C_6$$

160. The required number of possible selection

$$= {}^5C_2 \times {}^6C_3$$

161. Total number of persons = 6 + 4 = 10

Hence, the total number of possible selection

$$= {}^{10}C_5 - {}^6C_5$$

$$= \frac{10!}{5! \times 5!} - 6$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6}{120} - 6$$

$$= 63 \times 4 - 6$$

$$= 252 - 6$$

$$= 246$$

162. Total number of children = 6 + 7 = 13

Hence, the possible number of selection

$$= {}^{13}C_5 - {}^6C_5$$

$$= \frac{13!}{5! \times 8!} - 6$$

$$= \frac{13 \times 12 \times 11 \times 10 \times 9}{120} - 6$$

$$= 13 \times 11 \times 9 - 6$$

$$= 1289 - 6$$

$$= 1283$$

163. The possible number of selections

$$= {}^6C_2 \times {}^6C_5 + {}^6C_3 \times {}^6C_4 + {}^6C_4 \times {}^6C_3 + {}^6C_5 \times {}^6C_2$$

$$= 15 \times 6 + 20 \times 15 + 15 \times 20 + 6 \times 15$$

$$= 90 + 300 + 300 + 90$$

$$= 780$$

164. The triplet of 7 considered as (2, 2, 3).

Hence, the possible number of selections

$$= {}^4C_2 \times {}^5C_2 \times {}^6C_3 + {}^4C_2 \times {}^5C_3 \times {}^6C_2 + {}^4C_3 \times {}^5C_2 \times {}^6C_2$$

$$= 6 \times 10 \times 20 + 6 \times 10 \times 15 + 4 \times 10 \times 15$$

$$= 1200 + 900 + 600$$

$$= 2700$$

165. The possible ways he can vote = ${}^6C_1 + {}^6C_2 + {}^6C_3$

$$= 6 + 15 + 20 = 41$$

166. Hence, the number of possible ways Master Amit can

$$\text{invite his friends} = {}^8C_1 + {}^8C_2 + {}^8C_3 + \dots + {}^8C_{10}$$

$$= 2^8 - 1$$

$$= 256 - 1$$

$$= 255$$

167. The number of possible ways Master Roshan can give the answer is $= {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + \dots + {}^{10}C_{10}$
 $= 2^{10} - 1$
 $= 1024 - 1$
 $= 1023$

168. The student will fail if he/she fails in one or more subjects.

The total number of ways he/she can fail

$$= {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$$

$$= 2^5 - 1$$

$$= 32 - 1 = 31$$

169. The number of possible ways, the student select the questions

$$= ({}^3C_1 + {}^3C_2 + {}^3C_3) \times ({}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5)$$

$$= (2^3 - 1) \times (2^5 - 1)$$

$$= 7 \times 31 = 217$$

170. (i) The number of possible selections, in which at least one cube is white, is

$$= {}^2C_1 \times {}^3C_2 + {}^2C_2 \times {}^3C_1$$

$$= 2 \times 3 + 1 \times 3$$

$$= 6 + 3$$

$$= 9$$

- (ii) The number of possible selections, in which at least one cube is red, is

$$= {}^3C_1 \times {}^3C_2 + {}^3C_2 \times {}^2C_1$$

$$= 3 \times 1 + 3 \times 2$$

$$= 3 + 6$$

$$= 9$$

171. The number of visits of teacher
 = Number of groups of 3 children out of n
 $= {}^nC_3$

Number of visits of a particular child $= {}^{n-1}C_2$

According to the given condition,

$${}^nC_3 - {}^{n-1}C_2 = 84$$

$$\Rightarrow \frac{n(n-1)(n-2)}{6} - \frac{(n-1)(n-2)}{2} = 84$$

$$\Rightarrow \frac{(n-1)(n-2)}{2} \left(\frac{n}{3} - 1 \right) = 84$$

$$\Rightarrow \frac{(n-1)(n-2)(n-3)}{6} = 84$$

$$\Rightarrow (n-1)(n-2)(n-3) = 504$$

$$\Rightarrow (n-1)(n-2)(n-3) = 9 \times 8 \times 7$$

$$\Rightarrow (n-1) = 9$$

$$\Rightarrow n = 10$$

172. Let the number of contestants be n .

According to the questions,

$${}^{n+1}C_1 + {}^{n+1}C_2 + {}^{n+1}C_3 + \dots + {}^{n+1}C_n = 126$$

$$\Rightarrow 2^{n+1} - 2 = 126$$

$$\Rightarrow 2^{n+1} = 128$$

$$\Rightarrow 2^{n+1} = 2^7$$

$$\Rightarrow n + 1 = 7$$

$$\Rightarrow n = 6.$$

Thus, the number of contestants = 6.

173. Total number of games played by the remaining players = $117 - 12 = 105$

Let the number of remaining participants be n .

Thus,

$${}^nC_2 = 105$$

$$\Rightarrow \frac{n(n-1)}{2} = 105$$

$$\Rightarrow n(n-1) = 210$$

$$\Rightarrow n^2 - n - 210 = 0$$

$$\Rightarrow (n-15)(n+14) = 0$$

$$\Rightarrow n = 15$$

Hence, the number of participants at the beginning

$$= 15 + 2 = 17$$

174. It is given that

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n = 256 \quad \dots(i)$$

$${}^{2n+1}C_{2n+1} + {}^{2n+1}C_{2n} + {}^{2n+1}C_{2n-1} + \dots + {}^{2n+1}C_{n+1} = 256 \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_{2n+1} = 512$$

$$\Rightarrow 2^{2n+1} = 512 = 2^9$$

$$\Rightarrow 2n + 1 = 9$$

$$\Rightarrow n = 4$$

175. It is given that

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n + {}^{2n+1}C_{n+1} \quad \dots(i)$$

$${}^{2n+1}C_{2n} + {}^{2n+1}C_{2n-1} + {}^{2n+1}C_{2n-2} + \dots + {}^{2n+1}C_{n+1} = 63 \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n + {}^{2n+1}C_{n+1} + \dots + {}^{2n+1}C_{2n-1} + {}^{2n+1}C_{2n} = 63 + 63$$

$$\Rightarrow 2^{2n+1} - 2 = 126$$

$$\Rightarrow 2^{2n+1} = 128$$

$$\Rightarrow 2^{2n+1} = 128 = 2^7$$

$$\Rightarrow 2n + 1 = 7$$

$$\Rightarrow n = 3$$

Hence, the value of n is 3.

176. Here, all balls are in red colours.

So, the objects are identical.

Hence, the total number of selections is 1.

177. Here, all balls are in green colours.

So, the objects are identical.

As we know that the number of selections of zero or more objects from n identical objects is $n + 1$.

Thus, the total number of possible selections = $12 + 1 = 13$.

178. The total number of possible selections

$$= (10 + 1)(15 + 1)(12 + 1)(5 + 1) - 1$$

$$= 11 \times 16 \times 13 \times 6 - 1$$

$$= 17678 - 1$$

$$= 13728 - 1$$

$$= 13727$$

179. The number of possible selections

$$\begin{aligned}
 &= (6+1)(5+1)(4+1) - 1 \\
 &= 7 \times 6 \times 5 - 1 \\
 &= 210 - 1 \\
 &= 209
 \end{aligned}$$

180. The number of possible selections

$$\begin{aligned}
 &= 6 \times 5 \times 7 \\
 &= 210
 \end{aligned}$$

181. The number of selections

$$\begin{aligned}
 &= (10+1) \times 6 \times (4+1) \\
 &= 11 \times 6 \times 5 \\
 &= 66 \times 5 \\
 &= 330
 \end{aligned}$$

182. The number of possible selections

$$\begin{aligned}
 &= (4+1)(5+1)2^3 - 1 \\
 &= 5 \times 6 \times 8 - 1 \\
 &= 240 - 1 \\
 &= 239
 \end{aligned}$$

183. Given word is 'ABRACABDABRA' in which 5 As, 2 Bs, 2 Rs, 1 C and 1 D.

Hence, the number of selections

$$\begin{aligned}
 &= (5+1)(2+1)(2+1)2^2 - 1 \\
 &= 54 \times 4 - 1 \\
 &= 216 - 1 \\
 &= 215
 \end{aligned}$$

184. Given word is MATHEMATICS in which 2 Ms, 2 As, 2 Ts and H, E, I, C, S are 1 each.

Hence, the number of selections

$$\begin{aligned}
 &= (2+1)(2+1)(2+1)2^5 - 1 \\
 &= 9 \times 32 - 1 \\
 &= 198 - 1 \\
 &= 197
 \end{aligned}$$

185. There are 9 Ds, 3 As, 2 Ys, 3 Es, 1 I and 1 L.

Hence, the number of selections

$$\begin{aligned}
 &= (9+1)(3+1)(2+1)(3+1)2^2 - 1 \\
 &= 10 \times 4 \times 3 \times 4 \times 4 - 1 \\
 &= 10 \times 192 - 1 \\
 &= 1920 - 1 \\
 &= 1919
 \end{aligned}$$

186. There are 10 letters of six different types, namely, O, O, O; P, P; R, R; T; I; N.

i.e. 3 Os, 2 Ps, 2 Rs, 1 T, 1 I, 1 N.

Case I: When 3 alike and 1 different,

the number of selections $= {}^1C_1 \times {}^5C_1 = 1 \times 5 = 5$

Case II: When 2 alike of one kind and another 2 alike of 2nd kind, the number of selections $= {}^3C_2 = 3$

Case III: When 2 are alike and another 2 are different, the number of selections $= {}^3C_1 \times {}^5C_2 = 3 \times 10 = 30$

Case IV: When all 4 are different, the number of selections

$$= {}^6C_4 = \frac{6 \times 5}{2} = 15$$

Therefore, the number of total selections

$$= 5 + 3 + 30 + 15 = 53.$$

187. The total number of possible arrangements

$$\begin{aligned}
 &= 5 \times \frac{4!}{3!} + 3 \times \frac{4!}{2!2!} + 30 \times \frac{4!}{2!} + 15 \times 4! \\
 &= 5 \times 4 + 3 \times 6 + 30 \times 12 + 15 \times 24 \\
 &= 20 + 18 + 360 + 360 \\
 &= 758
 \end{aligned}$$

188. There are 11 letters of 8 different types, namely, MM, AA, TT, H, E, I, C, S, i.e. 2 Ms, 2 As, 2 Ts, 1 H, 1 E, 1 I, 1 C, 1 S.

Case I: When 2 alike of one kind and another 2 are alike of 2nd kind, the number of selections $= {}^3C_2 = 3$

Case II: When 2 are alike and 2 are different, the number of selections

$$= {}^3C_1 \times {}^7C_2 = 3 \times \frac{7 \times 6}{2} = 63$$

Case III: When all 4 are different, the number of selections

$$= {}^8C_4 = \frac{8!}{4! \times 4!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{24} = 70$$

Therefore, the number of total selections

$$\begin{aligned}
 &= 3 + 63 + 70 \\
 &= 136
 \end{aligned}$$

189. Total number of possible arrangements

$$\begin{aligned}
 &= 3 \times \frac{4!}{2!2!} + 63 \times \frac{4!}{2!} + 70 \times 4! \\
 &= 3 \times 6 + 63 \times 12 + 70 \times 24 \\
 &= 18 + 7560 + 1680 \\
 &= 2454
 \end{aligned}$$

190. There are 8 letters of 6 different types, namely, PP, SS, A, O, R, T, i.e. 2 Ps, 2 Ss, 1 A, 1 O, 1 R, 1 T.

Case I: When 2 alike of one kind and another 2 are alike of 2nd kind, the number of selections $= {}^2C_2 = 1$

Case II: When 2 are alike and 2 are different, the number of selections

$$= {}^2C_1 \times {}^5C_1 = 2 \times 5 = 10$$

Case III: When all 4 are different, the number of selections

$$= {}^6C_4 = \frac{6 \times 5}{2} = 15$$

Therefore, the total number of selections

$$= 1 + 10 + 15 = 26$$

191. Total number of possible arrangements

$$\begin{aligned}
 &= 1 \times \frac{4!}{2!2!} + 10 \times \frac{4!}{2!} + 15 \times 4! \\
 &= 1 \times 6 + 10 \times 12 + 15 \times 24 \\
 &= 6 + 120 + 360 \\
 &= 486.
 \end{aligned}$$

192. There are 14 letters of 8 different types, namely, N, N, N; A, A; I, I; T, T; E, O; L, R, i.e. 3 Ns, 2 As, 2 Is, 2 Ts, 1 E, 1 O, 1 L, 1 R.

Case I: When 3 are alike and 2 are different, the number of selections

$$= {}^1C_1 \times {}^7C_2 = 1 \times 21 = 21$$

Case II: When 3 are alike of one kind and 2 are alike of 2nd kind, the number of selections

$$= {}^1C_1 \times {}^3C_1 = 1 \times 3 = 3$$

Case III: When 2 are alike and 3 are different, the number of selections

$$= {}^4C_1 \times {}^7C_3 = 4 \times 35 = 140$$

Case IV: When 2 sets are alike and 1 is different, the number of selections

$$= {}^4C_2 \times {}^6C_1 = 6 \times 6 = 36$$

Case V: When all are different, the number of selections

$$= {}^8C_5 = \frac{8 \times 7 \times 6}{6} = 56$$

Therefore, total number of selections

$$= 21 + 3 + 140 + 36 + 56 = 256$$

193. Total number of possible arrangements

$$= 21 \times \frac{5!}{3!} + 3 \times \frac{5!}{3!2!} + 140 \times \frac{5!}{2!} + 36 \times \frac{5!}{2!2!} + 56 \times 5!$$

$$= 21 \times 20 + 3 \times 10 + 140 \times 60 + 36 \times 30 + 561 \times 20$$

$$= 420 + 30 + 8400 + 1080 + 6720$$

$$= 450 + 9480 + 6720$$

$$= 16650$$

194. We have $\alpha = 10800 = 2^4 \times 3^3 \times 5^2$

(i) The total number of divisors

$$= (4+1)(3+1)(2+1)$$

$$= 5 \times 4 \times 3 = 60$$

(ii) The number of even divisors

$$= 4 \times (3+1) \times (2+1)$$

$$= 4 \times 4 \times 3$$

$$= 48$$

(iii) The number of divisors of the form $4m+2$ is

$$= \text{the number is of the form } 6, 10$$

$$= (3+1) \times (2+1)$$

$$= 12$$

(iv) Now, $10800 = 2^4 \times 3^3 \times 5^2$

$$= (3 \times 5)^2 \times 2^4 \times 3^1$$

Thus, the number of divisors of multiple of 15 is

$$= (2+1) \times (4+1) \times (1+1)$$

$$= 3 \times 5 \times 2$$

$$= 30$$

195. We have $18900 = 2^2 \times 3^3 \times 5^2 \times 7$

Hence, the required number of possible ways

$$= 2^{4-1} = 2^3 = 8$$

196. We have $2520 = 2^3 \times 3^2 \times 5 \times 7$

Thus, the total number of possible ordered pairs (p, q) is

$$= (2.3 - 1) \times (2.2 + 1) \times (2.1 + 1) \times (2.1 + 1)$$

$$= 7 \times 5 \times 3 \times 3$$

$$= 315$$

197. We have $2520 = 2^3 \times 3^2 \times 5 \times 7$

Hence, the sum of all even divisors of 2520

$$= (1 + 2 + 2^2 + 2^3) \times (1 + 3 + 3^2) \times (1 + 5) \times (1 + 7)$$

$$= 15 \times 13 \times 6 \times 8$$

$$= 9360$$

198. We have $360 = 2^3 \times 3^2 \times 5$

Hence, the sum of all odd divisors of 360

$$= (1 + 3 + 3^2) \times (1 + 5)$$

$$= 13 \times 6$$

$$= 78$$

199. We have $38808 = 2^3 \times 3^2 \times 7^2 \times 1$

Hence, the sum of all even divisors

$$= 1 + 2 + 2^2 + 2^3$$

$$= 15$$

200. We have $7200 = 2^5 \times 3^2 \times 5^2$

Since the sum of all odd factors that ends with 5 is an odd factor, so all the factors will be $3^2 \times 5^2$.

Hence, the required sum of all factors ends with 5

$$= (1 + 3 + 3^2) \times (1 + 5 + 5^2)$$

$$= 13 \times 31$$

$$= 403$$

201. The numbers $1, p, p^2$ has 3 factors only when p is a prime number.

As we know that, $100^2 = 10000$

Thus, there are 25 prime numbers that exist under 100.

Hence, the required number = 100.

202. We have, $8100 = 2^2 \times 3^4 \times 5^2 = \text{a perfect square}$

Hence, the number of possible ways

$$= \frac{1}{2}[(2+1)(4+1)(2+1) + 1]$$

$$= 23$$

203. We have, $10800 = 2^4 \times 3^3 \times 5^2 \neq \text{a perfect square}$.

Hence, the number of possible ways

$$= \frac{1}{2}[(4+1) \times (3+1) \times (2+1)]$$

$$= 30$$

204. Set I Set II No. of ways

P Q, R, S

Q P, R, S ${}^4C_1 \times {}^4C_3$

R P, Q, S

S P, Q, R

Hence, the number of possible ways

$$= {}^4C_1 \times {}^4C_3$$

$$= 4$$

205. Person I Person II No. of ways

P Q, R, S

Q P, R, S

R P, Q, S

S P, Q, R ${}^4C_3 \times {}^1C_1 \times 2!$

Q, R, S P
P, R, S Q
P, Q, S R
P, Q, R S

Hence, the number of possible ways

$$= {}^4C_3 \times {}^1C_1 \times 2!$$

$$= 4 \times 2 = 8$$

206. Set I Set II No. of ways

P, Q R, S
P, R Q, S

P, S Q, R
Q, R P, S
Q, S P, R
R, S P, Q

$$\frac{{}^4C_3 \times {}^1C_1}{2!}$$

Hence, the number of possible ways

$$= \frac{{}^4C_3 \times {}^1C_1}{2!}$$

$$= 3$$

207. Person I Person II No. of ways

P, Q R, S
P, R Q, S
P, S Q, R
Q, R P, S
Q, S P, R
R, S P, Q

$${}^4C_2 \times {}^2C_2$$

Hence, the number of possible ways

$$= {}^4C_2 \times {}^2C_2$$

$$= 6$$

208. The possible triplets of 5 are

(1, 1, 3) and (1, 2, 2).

Hence, the number of possible ways

$$= 5! \times \frac{3!}{2!} + 5! \times \frac{3!}{2!}$$

$$= 120 \times 3 + 120 \times 3$$

$$= 360 + 360$$

$$= 720$$

Alternate method

The number of possible ways

$$= {}^{5-1}C_{3-1} \times 5!$$

$$= {}^4C_2 \times 5!$$

$$= 6 \times 120$$

$$= 720$$

209. The possible triplets of 5 are

(1, 1, 3) and (1, 2, 2).

Hence, the number of possible ways

$$= \frac{5!}{1!1!3!} \times 3! + \frac{5!}{2!2!1!} \times 3!$$

$$= 120 + 180$$

$$= 300$$

210. The possible doublets of 4 are (1, 3) and (2, 2).

Hence, the number of possible ways

$$= \frac{4!}{1!3!} \times 2! + \frac{4!}{2!2!} \times \frac{2!}{2!}$$

$$= \frac{4!}{3} + \frac{4!}{2!2!}$$

$$= \frac{24}{3} + \frac{24}{4}$$

$$= 8 + 6$$

$$= 14$$

211. The possible triplets of 5 are (1, 1, 3) and (1, 2, 2).

Hence, the number of possible ways

$$= \frac{5!}{1!1!3!} \times \frac{3!}{2!} + \frac{5!}{2!2!1!} \times \frac{3!}{2!}$$

$$= \frac{5!}{2!} + \frac{5! \times 3!}{2!2!2!}$$

$$= 60 + 90$$

$$= 150$$

212. The possible triplets are

(1, 2, 3), (1, 1, 4) and (2, 2, 2).

Hence, the number of possible ways

$$= \frac{6!}{1!2!3!} \times 3! + \frac{6!}{1!1!4!} \times \frac{3!}{2!} + \frac{6!}{2!2!2!} \times \frac{3!}{3!}$$

$$= \frac{6!}{2!} + \frac{6!}{4 \times 2!} + \frac{6!}{2!2!2!}$$

$$= \frac{720}{2} + \frac{720}{8} + \frac{720}{8}$$

$$= 360 + 90 + 90$$

$$= 540$$

213. The number of possible ways

$$= {}^{10-1}C_{3-1}$$

$$= {}^9C_2$$

$$= \frac{9 \times 8}{2}$$

$$= 36$$

Alternate method

Here, the possible triplets are

(1, 3, 6), (1, 4, 5), (1, 2, 7), (2, 3, 5), (1, 1, 8), (2, 2, 6), (2, 4, 4) and (3, 3, 4)

Hence, the number of possible ways

$$= (1 \times 3! + 1 \times 3! + 1 \times 3! + 1 \times 3!)$$

$$+ \left(1 \times \frac{3!}{2!} + 1 \times \frac{3!}{2!} + 1 \times \frac{3!}{2!} + 1 \times \frac{3!}{2!} \right)$$

$$= 6 \times 4 + 3 \times 4$$

$$= 24 + 12$$

$$= 36$$

214. The number of possible ways

$$= {}^{5-1}C_{3-1}$$

$$= {}^4C_2 = \frac{4 \times 3}{2} = 6$$

Alternate method

Here, the possible triplets are

(1, 1, 3) and (1, 2, 2).

Hence, the number of possible ways

$$= 1 \times \frac{3!}{2!} + 1 \times \frac{3!}{2!}$$

$$= 3 + 3$$

$$= 6$$

215. The number of possible ways

$$= {}^{4-1}C_{2-1}$$

$$= {}^3C_1 = 3$$

Alternate method

Here, the possible doublets are

(1, 3) and (2, 2).

Hence, the number of possible ways

$$= 1! \times 2! + 1! \times \frac{2!}{2!}$$

$$= 2 + 1$$

$$= 3$$

216. The number of possible ways

$$= {}^{20+4-1}C_{4-1}$$

$$= {}^{23}C_3$$

$$= \frac{23 \times 22 \times 21}{6}$$

$$= 23 \times 11 \times 7$$

$$= 253 \times 7$$

$$= 1771$$

217. The number of possible selections is

$$= \sum_{k=0}^{12} {}^{25-10}C_k$$

$$= \sum_{k=0}^{12} {}^{15}C_k$$

$$= {}^{15}C_0 + {}^{15}C_1 + {}^{15}C_2 + \dots + {}^{15}C_{12}$$

$$= 2^{15} - ({}^{15}C_{13} + {}^{15}C_{14} + {}^{15}C_{15})$$

$$= 2^{15} - (105 + 15 + 1)$$

$$= 2^{15} - 121$$

218. (i) The number of possible selections

$$= \sum_{k=8-6}^8 {}^9C_k$$

$$= \sum_{k=2}^8 {}^9C_k$$

$$= {}^9C_2 + {}^9C_3 + {}^9C_4 + {}^9C_5 + {}^9C_6 + {}^9C_7 + {}^9C_8$$

$$= 29 - ({}^9C_0 + {}^9C_1 + {}^9C_9)$$

$$= 512 - 11$$

$$= 501$$

(ii) The number of possible selections

$$\sum_{k=0}^4 {}^9C_k = {}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + {}^9C_4$$

$$= 1 + 9 + 36 + 84 + 126$$

$$= 256$$

219. The number of possible ways

$$= 2! \times \left(1 - \frac{1}{1!} + \frac{1}{2!}\right)$$

$$= 1$$

220. The number of possible ways

$$= 3! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right)$$

$$= 3! \times \left(\frac{1}{2!} - \frac{1}{3!}\right)$$

$$= 3! \times \left(\frac{3-1}{3!}\right)$$

$$= 2$$

221. The number of possible ways

$$= 4! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right)$$

$$= 4! \times \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right)$$

$$= 4! \times \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right)$$

$$= 4! \times \frac{12-4+1}{24}$$

$$= 9$$

222. The number of possible ways

$$= 5! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 5! \times \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 5! \times \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right)$$

$$= 5! \times \left(\frac{60-20+5-1}{120}\right)$$

$$= 44$$

223. The number of total functions

$$= 5! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 44$$

224. First of all, to distribute 5 right-handed shocks to 5 persons P_1, P_2, P_3, P_4 and P_5 respectively.

This can be done in $5!$ ways and for each distribution of right-handed shocks, left-handed shocks can be deranged in $D_5 = 44$ ways.

Hence, the number of possible ways

$$\begin{aligned}
 &= 5! \times 4 \\
 &= 1320 \times 4 \\
 &= 5280
 \end{aligned}$$

225. 2 letters can be selected out of 5 letters in 5C_2 ways. For each such selection, the remaining 3 letters placed in wrong envelopes in D_3 ways.

Hence, the number of possible selections

$$\begin{aligned}
 &= {}^5C_2 \times D_3 \\
 &= 10 \times 2 \\
 &= 20
 \end{aligned}$$

226. The number of ways to place at least 2 balls in boxes of the same color

= The number of ways without restriction – the number of ways all 5 balls are wrongly placed – the number of ways to select a ball in right box $\times$ number of ways in which remaining 4 balls in 4 wrong boxes

$$\begin{aligned}
 &= 5! - 44 - {}^5C_1 \times D_4 \\
 &= 120 - 44 - 5 \times 9 \\
 &= 120 - 44 - 45 \\
 &= 120 - 89 \\
 &= 31
 \end{aligned}$$

227. The number of possible ways

$$\begin{aligned}
 &= 0 \text{ letter in wrong envelope and 6 are correct} \\
 &\quad + 1 \text{ letter in wrong envelope and 5 are correct} \\
 &\quad + 2 \text{ letters in wrong envelopes and 4 are correct} \\
 &\quad + 3 \text{ letters in wrong envelopes and 3 are correct} \\
 &= 1 + 0 \text{ (not possible)} + {}^6C_4 \times D_2 + {}^6C_3 \times D_3 \\
 &= 1 + \frac{6 \times 5}{2} \times 1 + \frac{6 \times 5 \times 4}{6} \times 2 \\
 &= 1 + 15 + 40 \\
 &= 56
 \end{aligned}$$

228. The number of possible ways = the number of ways pick up 1 right coat and 5 wrong coats

$$\begin{aligned}
 &= {}^6C_1 \times D_5 \\
 &= 6 \times 44 \\
 &= 264
 \end{aligned}$$

229. There are 11 letters in which 3 Es, 3 Ns, 2 Gs, 2 Is and 1 R.

$$\begin{aligned}
 &= \text{Co-efficient of } x^4 \text{ in} \\
 &\quad (1 + x + x^2 + x^3)^2 \times (1 + x + x^2) \times (1 + x)^3
 \end{aligned}$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$\left(\frac{1-x^4}{1-x} \right)^2 \times \left(\frac{1-x^3}{1-x} \right)^2 \times (1+x)$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$\{(1-x^4)(1-x^3)\}^2 (1+x)(1-x)^{-4}$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$(1-x^3-x^4)(1+x)(1-x)^{-4}$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$(1-x-2x^3-4x^4) (1 + {}^4C_1x + {}^5C_2x^2 + {}^6C_3x^3 + {}^7C_4x^4)$$

$$= {}^7C_4 + {}^6C_3 - 4 - 4 \cdot {}^2C_1$$

$$= 35 + 20 - 4 - 8$$

$$= 43$$

The number of permutations of 4 letters is = Co-efficient of x^4 in

$$4! \times \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} \right)^2 \times \left(1 + \frac{x}{1!} + \frac{x^2}{2!} \right)^2 \times (1+x)$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$4! \times \left(1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} \right)^2 \times \left(1 + \frac{x}{1} + \frac{x^2}{2} \right)^2 \times (1+x)$$

$$= \text{Co-efficient of } x^4 \text{ in}$$

$$4! \times \left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{7}{12}x^4 \right)$$

$$\times \left(1 + 3x + 4x^2 + 3x^3 + \frac{5}{4}x^4 \right)$$

$$= 4! \times \left(\frac{7}{12} + \frac{5}{4} + (6+4) + 8 \right)$$

$$= 24 \times \frac{22}{12} + 24 \times 18$$

$$= 44 + 432$$

$$= 476$$

230. There are 11 letters in which 3 Es, 3 Ns, 2 Ds, 1 I, 1 P and 1 T, respectively.

Hence, the total number of selections

$$= \text{Co-efficient of } x^5 \text{ in } (1+x+x^2+x^3)^2 \times (1+x+x^2) \times (1+x)^3$$

$$= \text{Co-efficient of } x^5 \text{ in}$$

$$\left(\frac{1-x^4}{1-x} \right)^2 \times \left(\frac{1-x^3}{1-x} \right) \times (1+x)^3$$

$$= \text{Co-efficient of } x^5 \text{ in}$$

$$(1-2x^4)(1-x^3) \times (1-x)^{-3} \times (1+x)^3$$

$$= \text{Co-efficient of } x^5 \text{ in}$$

$$(1-x^3-2x^4) \times (1+3x+3x^2+x^3) \times (1-x)^{-3}$$

$$= \text{Co-efficient of } x^5 \text{ in}$$

$$(1+3x+3x^2-5x^4-9x^5)$$

$$(1 + {}^3C_1x + {}^4C_2x^2 + {}^5C_3x^3 + {}^6C_4x^4 + {}^7C_5x^5)$$

$$= ({}^7C_5 - 9) + (3 \times {}^6C_4 - 15) + (3 \times {}^5C_3)$$

$$= 21 - 9 + 45 - 15 + 30$$

$$= 12 + 30 + 30$$

$$= 72$$

231. The number of possible ways

$$= \text{Co-efficient of } x^{15} \text{ in } (x + x^2 + \dots + x^6)^3$$

$$= \text{Co-efficient of } x^{12} \text{ in } (x + \dots + x^5)^3$$

$$= \text{Co-efficient of } x^{12} \text{ in } \left(\frac{1-x^6}{1-x} \right)^3$$

$$= \text{Co-efficient of } x^{12} \text{ in } (1-x^6)^3 \times (1-x)^{-3}$$

$$= \text{Co-efficient of } x^{12} \text{ in } (1-3x^6+3x^{12})$$

$$(1 + {}^3C_1x^6 + {}^4C_2x^{12} + \dots + {}^8C_6x^6 + \dots + {}^{14}C_{12}x^{12})$$

$$\begin{aligned}
 &= {}^{14}C_{12} - 3 \times {}^8C_6 + 3 \\
 &= 91 - 84 + 3 \\
 &= 10
 \end{aligned}$$

Alternate method

The number of possible triplets to get 15 are
(5, 5, 5), (5, 6, 4) and (6, 6, 4)

Thus, the number of possible ways

$$\begin{aligned}
 &= \frac{3!}{3!} + 3! + \frac{3!}{2!} \\
 &= 1 + 6 + 3 \\
 &= 10
 \end{aligned}$$

232. The number of possible ways

$$\begin{aligned}
 &= \text{Co-efficient of } x^{16} \text{ in } (x^3 + x^4 + \dots + x^7)^4 \\
 &= \text{Co-efficient of } x^4 \text{ in } (1 + x + x^2 + x^3 + x^4)^4 \\
 &= \text{Co-efficient of } x^4 \text{ in } \left(\frac{1-x^5}{1-x} \right)^5 \\
 &= \text{Co-efficient of } x^4 \text{ in } (1-x^5)^4 \times (1-x)^{-4} \\
 &= \text{Co-efficient of } x^4 \text{ in } (1-x)^{-4} \\
 &= \text{Co-efficient of } x^4 \text{ in } (1 + {}^4C_1x + {}^5C_2x^2 + {}^6C_3x^3 + {}^7C_4x^4 + \dots) \\
 &= {}^7C_4 \\
 &= 35
 \end{aligned}$$

Alternate method

The number of possible quadruplets are

(3, 3, 3, 7), (3, 3, 5, 5), (3, 3, 4, 6), (3, 4, 4, 5) and (4, 4, 4, 4).

Hence, the number of possible ways

$$\begin{aligned}
 &= \frac{4!}{3!} + \frac{4!}{2!2!} + \frac{4!}{2!} + \frac{4!}{2!} + \frac{4!}{2!} \\
 &= 4 + 6 + 12 + 12 + 1 \\
 &= 35
 \end{aligned}$$

233. (i) The total number of positive solutions

$$\begin{aligned}
 &= {}^{20+4-1}C_{4-1} \\
 &= {}^{23}C_3 \\
 &= \frac{23.22.21}{6} \\
 &= 1771
 \end{aligned}$$

(ii) Here, $x \geq 1, y \geq 1, z \geq 1, w \geq 1$

$$\Rightarrow x-1 \geq 0, y-1 \geq 0, z-1 \geq 0, w-1 \geq 0$$

$$\text{Let } p = x-1, q = y-1, r = z-1, s = w-1$$

$$\text{Now, } x+y+z+w=20$$

$$\Rightarrow p+q+r+s+4=20$$

$$\Rightarrow p+q+r+s=16$$

Hence, the number of required solutions

$$\begin{aligned}
 &= {}^{16+4-1}C_{4-1} \\
 &= {}^{19}C_3 \\
 &= \frac{19.18.17}{6} \\
 &= 969
 \end{aligned}$$

234. Given $3x + y + z = 24, x \geq 0, y \geq 0, z \geq 0$

$$\text{Let } z = k.$$

$$\text{Then } y + z = 24 - 3k$$

$$\text{So, } 0 \leq 24 - 3k \leq 24$$

$$\Rightarrow -24 \leq -3k \leq 24$$

$$\Rightarrow 0 \leq k \leq 8$$

Thus, the number of integral solutions

$$\begin{aligned}
 &= {}^{24-3k+2-1}C_{2-1} \\
 &= {}^{25-3k}C_1 = (25 - 3k)
 \end{aligned}$$

Hence, the number of integral solutions

$$\begin{aligned}
 &= \sum_{k=0}^8 (25 - 3k) \\
 &= \sum_{k=0}^8 25 - 3 \sum_{k=0}^8 k \\
 &= 25 \times 9 - \frac{3 \times 8 \times 9}{2} \\
 &= 225 - 108 \\
 &= 117
 \end{aligned}$$

235. We have $2x + 2y + z = 10$

$$\Rightarrow x + y = \frac{10-z}{2}$$

$$\text{So, } 0 \leq x + y \leq 10$$

$$\text{Let } x + y + t = 10, x \geq 0, y \geq 0, z \geq 0$$

Hence, the number of non-negative integral solutions

$$\begin{aligned}
 &= {}^{20+4-1}C_{4-1} \\
 &= {}^{23}C_3 \\
 &= \frac{23.22}{2} = 66
 \end{aligned}$$

236. Here, $x-1 \geq 0, y-2 \geq 0, z-3 \geq 0, t-4 \geq 0$

$$\text{Let } p = x-1, q = y-2, r = z-3, s = t-4$$

$$\Rightarrow x = p+1, y = q+2, z = r+3, t = s+4$$

$$\text{Now, } x+y+z+t=29$$

$$\Rightarrow p+q+r+s=29-10=19$$

Hence, the number of non-negative solutions

$$\begin{aligned}
 &= {}^{19+4-1}C_{4-1} \\
 &= {}^{22}C_3 \\
 &= \frac{22.21.20}{6} \\
 &= 1540
 \end{aligned}$$

237. Given system of equations are

$$x + y + z - t + u = 20, x + y + z = 5$$

$$x + y + z = 5 \quad \dots(i)$$

$$\Rightarrow t + u = 15 \quad \dots(ii)$$

Now, the number of non-negative integral solutions of

$$\begin{aligned}
 (i) &= {}^{5+3-1}C_{3-1} \\
 &= {}^7C_2 = 21
 \end{aligned}$$

Also, the number of non-negative integral solutions of

$$\begin{aligned}
 (ii) &= {}^{15+2-1}C_{2-1} \\
 &= {}^{16}C_1 = 16
 \end{aligned}$$

Hence, the number of total non-negative integral solutions = $21 \times 16 = 336$

238. The total numbers of ordered triplets

$$= {}^{50}C_3 - {}^{19}C_3$$

239. Let $x = -p, y = -q, z = -r$, where $p, q, r > 0$

$$\text{Then } -p - q - r + 12 = 0$$

$$p + q + r = 12$$

Hence, the required number of points in space

$$= {}^{12-1}C_{3-1}$$

$$= {}^{11}C_2$$

$$= \frac{11 \times 10}{2} = 55$$

240. We have $x_1 x_2 x_3 = 30 = 2 \times 3 \times 5$

Hence, the number of positive integral solutions

$$= {}^{3+1-1}C_1 \times {}^{3+1-1}C_1 \times {}^{3+1-1}C_1$$

$$= {}^3C_1 \times {}^3C_1 \times {}^3C_1$$

$$= 3 \times 3 \times 3$$

$$= 27$$

241. We have $x_1 x_2 x_3 x_4 = 210$

$$= 2 \times 3 \times 5 \times 7$$

Hence, the number of positive integral solutions

$$= ({}^{4+1-1}C_1 \times {}^{4+1-1}C_1 \times {}^{4+1-1}C_1 \times {}^{4+1-1}C_1) \times (1 + {}^4C_2 + {}^4C_4)$$

(since all are positive or any two are negative or all four are negative)

$$= ({}^4C_1 \times {}^4C_1 \times {}^4C_1 \times {}^4C_1) \times (1 + {}^4C_2 + {}^4C_4)$$

$$= (4 \times 4 \times 4 \times 4) \times (1 + 6 + 1)$$

$$= 256 \times 8$$

$$= 2048$$

242. Here, the possible triplets of 12 are (1, 5, 6), (2, 4, 6), (2, 5, 5), (3, 4, 5) (3, 3, 6) and (4, 4, 4)

Hence, the number of possible ways to get a sum 12

$$= 3! + 3! + \frac{3!}{2!} + 3! + \frac{3!}{2!} + \frac{3!}{3!}$$

$$= 6 + 6 + 3 + 6 + 3 + 1$$

$$= 25$$

243. Let four boys be x_1, x_2, x_3 and x_4 .

$$\text{Thus, } x_1 + x_2 + x_3 + x_4 = 10,$$

where

$$x_1 \geq 1, x_2 \geq 1, x_3 \geq 1, x_4 \geq 1$$

$$\text{Let } p = x_1 - 1, q = x_2 - 1, r = x_3 - 1, s = x_4 - 1$$

$$\Rightarrow x_1 = p + 1, x_2 = q + 1, x_3 = r + 1, x_4 = s + 1$$

$$\text{Now, } x_1 + x_2 + x_3 + x_4 = 10$$

$$p + q + r + s + 4 = 0$$

$$p + q + r + s = 6$$

Hence, the number of possible ways

$$= \text{Number of non-negative solutions of the equation } p + q + r + s = 6$$

$$= {}^{6+4-1}C_{4-1}$$

$$= {}^9C_3$$

$$= \frac{9 \times 8 \times 7}{6}$$

$$= 84$$

Alternate method

Here, the possible quadruplets of 10 are

(1, 2, 3, 4), (1, 1, 4, 4), (2, 2, 2, 4), (1, 3, 3, 3), (1, 2, 5), (1, 1, 3, 5), (1, 1, 2, 6) and (1, 1, 1, 7).

Hence, the number of possible ways

$$= 4! + \frac{4!}{2!} + \frac{4!}{3!} + \frac{4!}{3!} + \frac{4!}{2!} + \frac{4!}{2!} + \frac{4!}{3!}$$

$$= 24 + 12 + 4 + 4 + 12 + 12 + 12 + 4$$

$$= 72 + 12$$

$$= 84$$

244. The number of possible selections = Co-efficient of in the expansion of

$$({}^2C_1x + {}^2C_2x^2) \times ({}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4) \\ ({}^2C_1x + {}^4C_2x^2)$$

= Co-efficient of x^6 in

$$({}^2C_1x + {}^2C_2x^2)^2 \times ({}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4)$$

= Co-efficient of x^6 in

$$(2x + x^2)^2 \times (4x + 6x^2 + 4x^3 + x^4)$$

= Co-efficient of x^6 in

$$(2 + x)^2 \times (4 + 6x + 4x^2 + x^3)$$

= Co-efficient of x^3 in

$$(2 + 4x + x^2) \times (4 + 6x + 4x^2 + x^3)$$

$$= 2 + 16 + 6$$

$$= 26$$

Alternate method

The number of possible selections

$$= {}^8C_6 - 2$$

$$= 28 - 2$$

$$= 26$$

Since either the first row or the third row may not have any x when the six x s are placed in the other two row

245. The number of arrangement of the letters P E R S O

$$= 6! = 720$$

Also, the number of selections of the letter of the word PERSON in the given square = Co-efficient of x^6 in the expansion of

$$({}^2C_1x + {}^2C_2x^2)^2 \times ({}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4) \\ = 26$$

Hence, the number of required possible ways

$$= 26 \times 6!$$

$$= 26 \times 720$$

$$= 18720$$

246. (i) The required number of straight lines

$$= {}^{12}C_2 - {}^5C_2 + 1$$

$$= 66 - 10 + 1$$

$$= 57$$

(ii) The number of required triangles

$$= {}^{12}C_3 - {}^5C_3$$

$$= 220 - 10$$

$$= 210$$

(iii) The number of quadrilaterals

$$= {}^5C_0 \times {}^7C_4 + {}^5C_1 \times {}^7C_3 + {}^5C_2 \times {}^7C_2$$

$$= 35 + 175 + 210$$

$$= 420$$

247. The number of diagonals

$$= {}^{10}C_2 - 10$$

$$= 45 - 10$$

$$= 35$$

248. Let the number of sides of the polygon be n .

Given ${}^nC_2 - 35$

$$\Rightarrow \frac{n(n-1)}{2} - n = 35$$

$$\Rightarrow n^2 - 3n - 70 = 0$$

$$\Rightarrow (n-10)(n+7) = 0$$

$$\Rightarrow n = 10$$

Hence, the number of sides of the polygon is 10.

249. The required number of triangles

= Total number of triangles – number of triangles having two sides common – number of triangles having one side common

$$= {}^8C_3 - 8 - {}^8C_1 \times {}^4C_1$$

$$= 56 - 8 - 32$$

$$= 56 - 40$$

$$= 16$$

250. Hence, the required number of parallelograms

$$= {}^{10}C_2 \times {}^8C_2$$

$$= 45 \times 8$$

$$= 1260$$

251. Given,

$$T_{n+1} - T_n = 21$$

$$\Rightarrow {}^{n+1}C_3 - {}^nC_3 = 21$$

$$\Rightarrow \frac{1}{6}[n(n+1)(n-1) - n(n-1)(n-2)] = 21$$

$$\Rightarrow n(n-1)[(n+1) - (n-2)] = 126$$

$$\Rightarrow 3n(n-1) = 126$$

$$\Rightarrow n(n-1) = 42$$

$$\Rightarrow n^2 - n - 42 = 0$$

$$\Rightarrow (n-7)(n+6) = 0$$

$$\Rightarrow n = 7$$

Hence, the value of n is 7.

252. The number of rectangles

$$= {}^{7-1}C_2 \times {}^{4-1}C_2$$

$$= {}^8C_2 \times {}^5C_2$$

$$= 28 \times 10$$

$$= 280$$

253. The number of squares

$$= 6 \times 9 + 5 \times 8 + 4 \times 7 + 3 \times 6 + 2 \times 5 + 1 \times 4$$

$$= 54 + 40 + 28 + 18 + 10 + 4$$

$$= 108 + 46$$

$$= 154$$

The number of rectangles

$$= {}^{9+1}C_2 \times {}^{6+1}C_2$$

$$= {}^{10}C_2 \times {}^7C_2$$

$$= 45 \times 21$$

$$= 945$$

Hence, the required number of rectangles excluding squares

$$= 945 - 154$$

$$= 791$$

254. The number of points of intersection of 5 circles

$$= 2 \times {}^5C_2$$

$$= 2 \times 10$$

$$= 20$$

255. The number of points of intersection of 8 lines and 4 circles

$$= {}^8C_2 + 2 \times {}^4C_2 + 2 \times {}^8C_1 \times {}^4C_1$$

$$= 28 + 12 + 64$$

$$= 104$$

256. Number of possible rectangles is

$$= [1 + 3 + 5 + \dots + (2m-1)] \times [1 + 3 + 5 + \dots + (2n-1)]$$

$$= m^2 n^2$$

257. Number of red lines = nC_2

Number of blue lines = n

Hence, ${}^nC_2 - n = n$

$$\Rightarrow {}^nC_2 = 2n$$

$$\Rightarrow \frac{n(n-1)}{2} = 2n$$

$$\Rightarrow n^2 - n = 4n$$

$$\Rightarrow n^2 = 5n$$

$$\Rightarrow n = 0, 5$$

Since $n \geq 2$, so $n = 5$.

258. The number of total triangles

$$= {}^3C_1 \times {}^4C_1 \times {}^5C_1 + \{ {}^3C_2 ({}^4C_1 + {}^5C_1) \}$$

$$+ {}^4C_2 ({}^3C_1 + {}^5C_1) + {}^5C_2 ({}^3C_1 + {}^4C_1) \}$$

$$= 60 + 3(4+5) + 6(3+5) + 10(3+4)$$

$$= 60 + 27 + 48 + 70$$

$$= 205$$

259. In 8×8 chess board, 7 pairs of adjacent columns be chosen in which first square be chosen in 2 ways (one from top and one from bottom) and 1 square be chosen from the 2nd column.

$$= 7(2 \times 1 + 6 \times 2)$$

$$= 7 \times 14$$

$$= 98$$